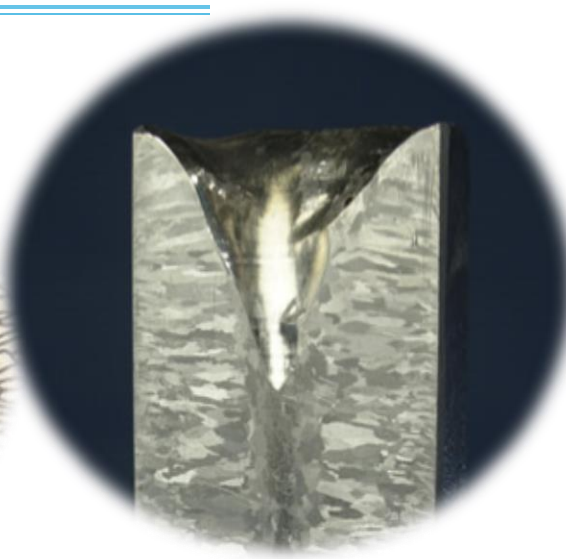
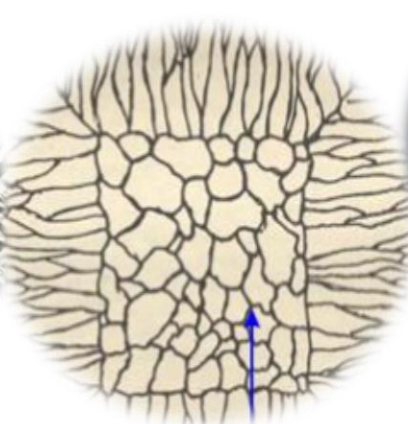
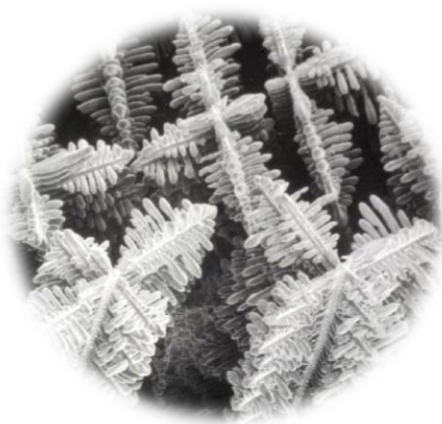


316-221 Metallurgy

Chapter 6: Solidification of Metals

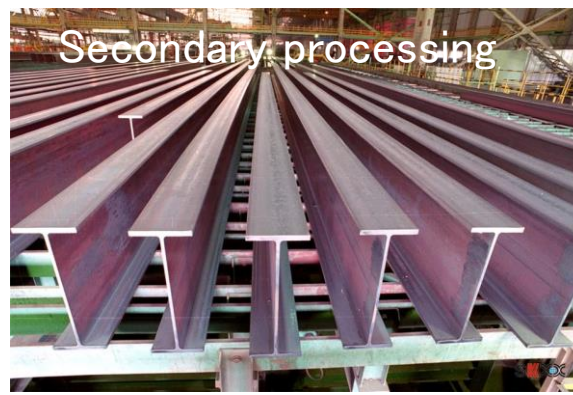


Contents

- Technological Significance
- Solidification of Pure Metals
- Cooling Curves
- Mechanism of Solidification of Metals
- Casting Processes
- Recovery, Recrystallization & Grain Growth

□ Technological Significance

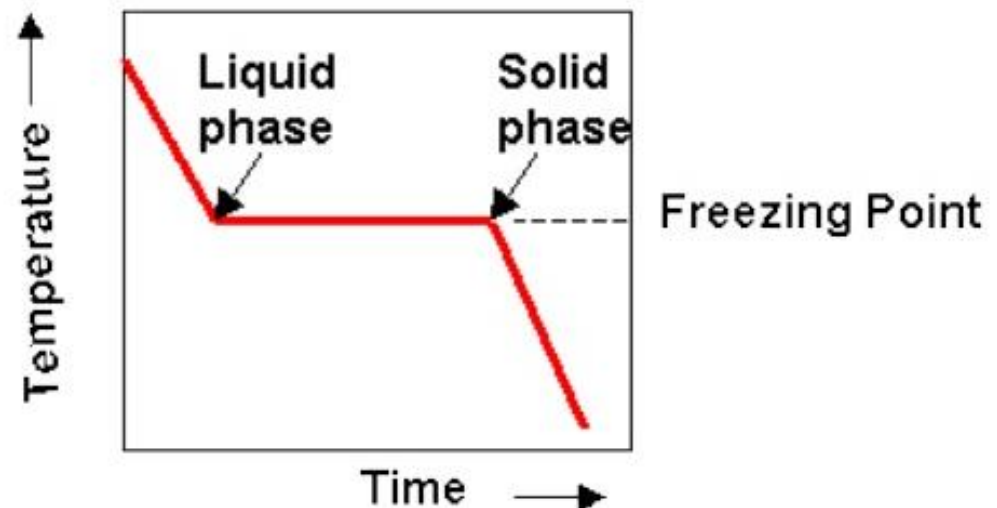
- **Primary processing** – Processes involving casting of molten metals into ingots or semi-finished useful shapes such as slabs.
- **Secondary processing** – Processes such as rolling, extrusion, etc. used to process ingots or slabs and other semi-finished shapes.



Solidification of Pure Metals

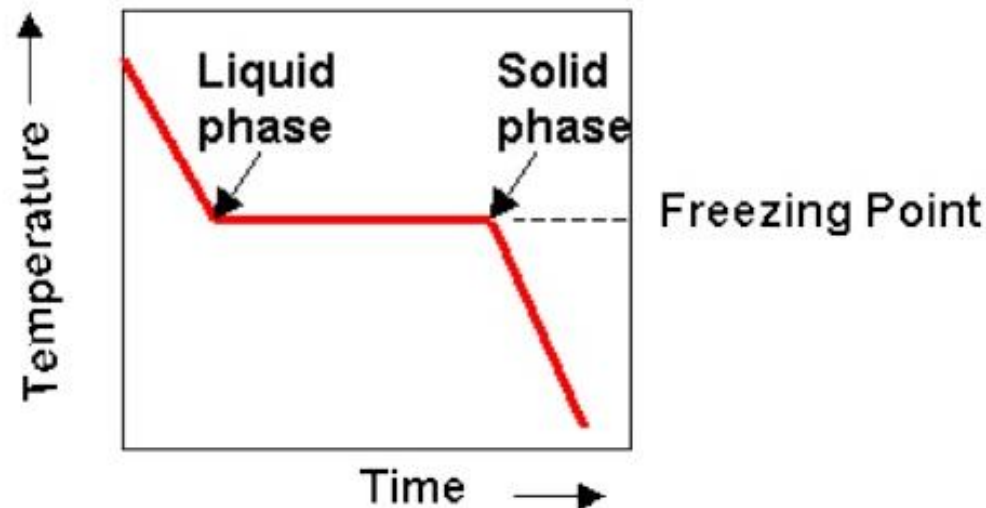
- ❑ During solidification, the liquid changes into solid during cooling.
- ❑ The energy of liquid is less than that of the solid above the melting point. Hence liquid stable above the melting point.
- ❑ Below the melting point, the energy of liquid become more than the of solid.
- ❑ Hence, below the melting point, the solid becomes more stable than the liquid.
- ❑ Therefore, at the melting point, liquid gets converted into the solid during cooling.

- ❑ This transformation of liquid into solid below melting point is know as **solidification**.



Solidification of Pure Metals

- ❑ Thermodynamically, both liquid and solid have equal energy at the melting point and therefore both are equally stable at melting point.
- ❑ Therefore, no solidification or melting will take place at the melting point. Liquid will remain liquid and solid will remain solid.
- ❑ Some under-cooling will be essential for solidification.
- ❑ This transformation occurs by **nucleation** and **growth**.



Solidification of Pure Metal at Equilibrium

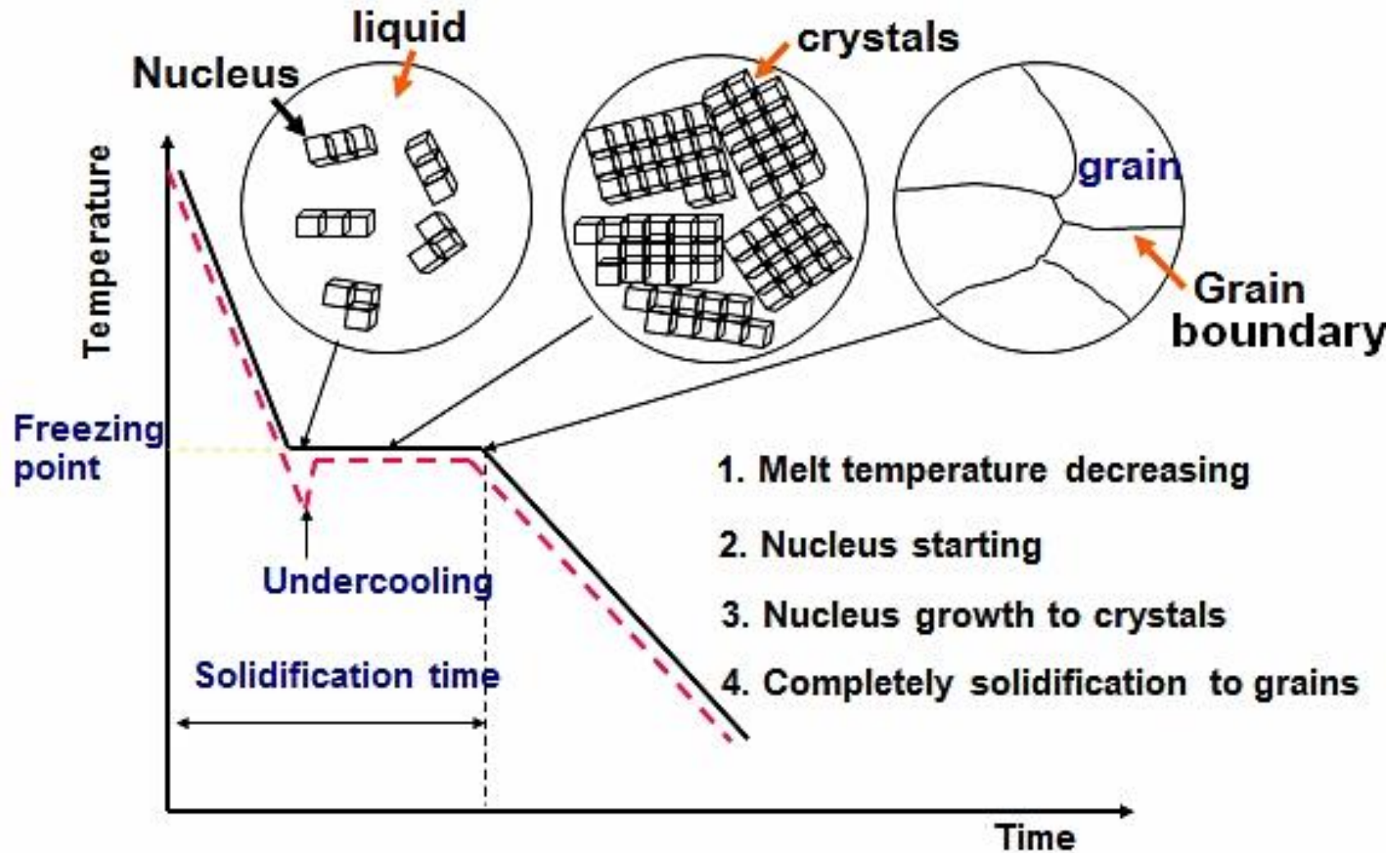
□ During the cooling process occurs simultaneously two processes

1. Nucleation

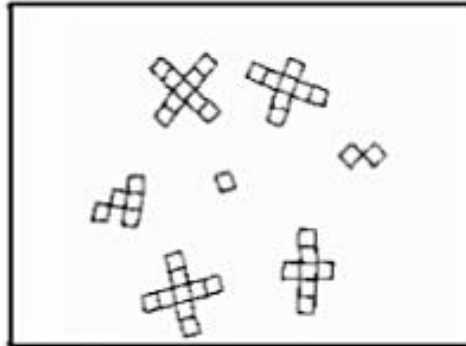
2. Crystal Growth

At equilibrium, transformation is slowly occurred and assumes a constant temperature and pressure, **no transfer of energy**.

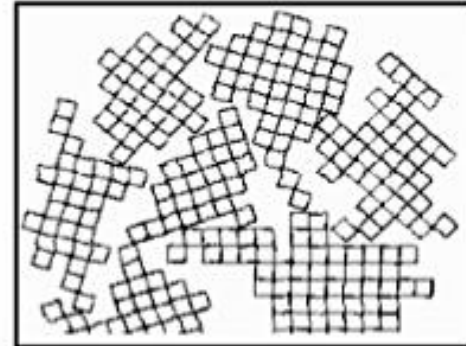
Solidification of Pure Metal at Equilibrium



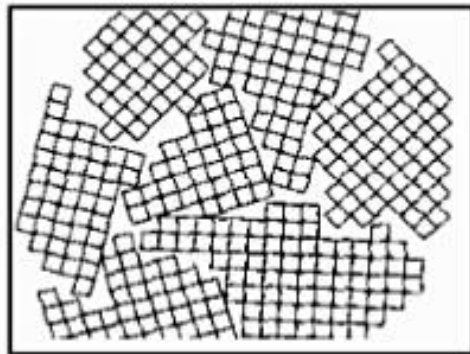
Stages during the solidification in pure metals



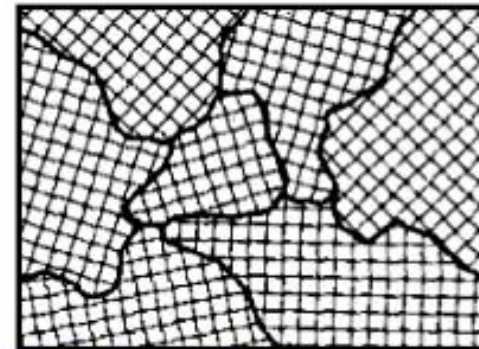
1. **Nucleation**: It begins at foreign particles in melt.



2. **Crystal growth**: Crystals begin to grow from each.



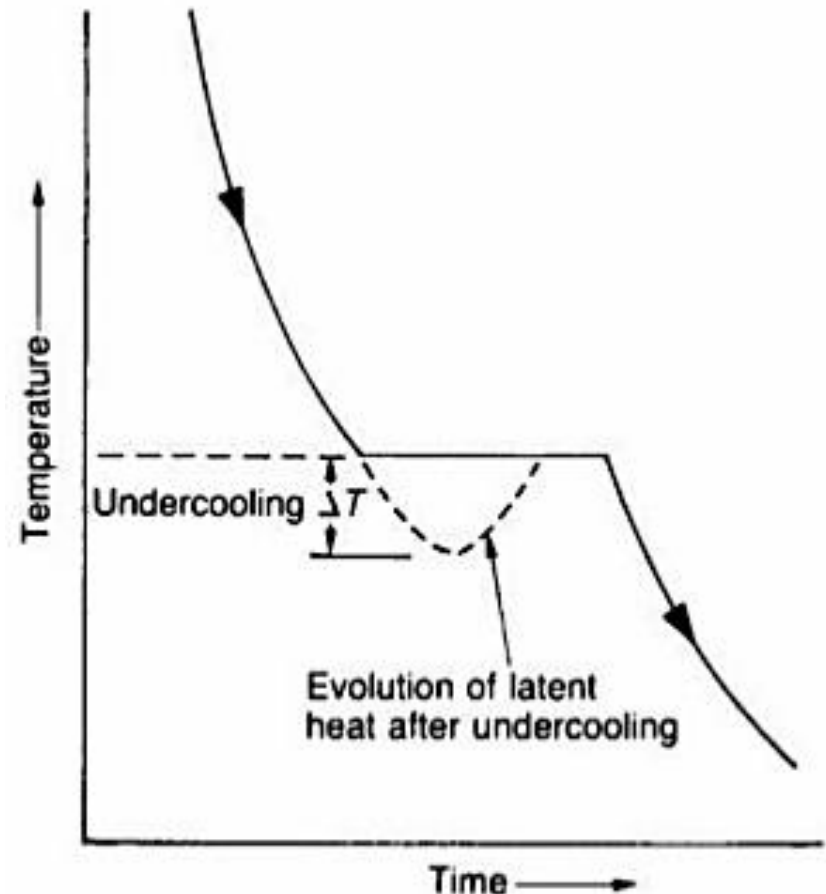
3. **Grain Formation**: Interface develops.



4. **Polycrystalline structure**: Grain growth is limited by another grain, creating a boundary between them

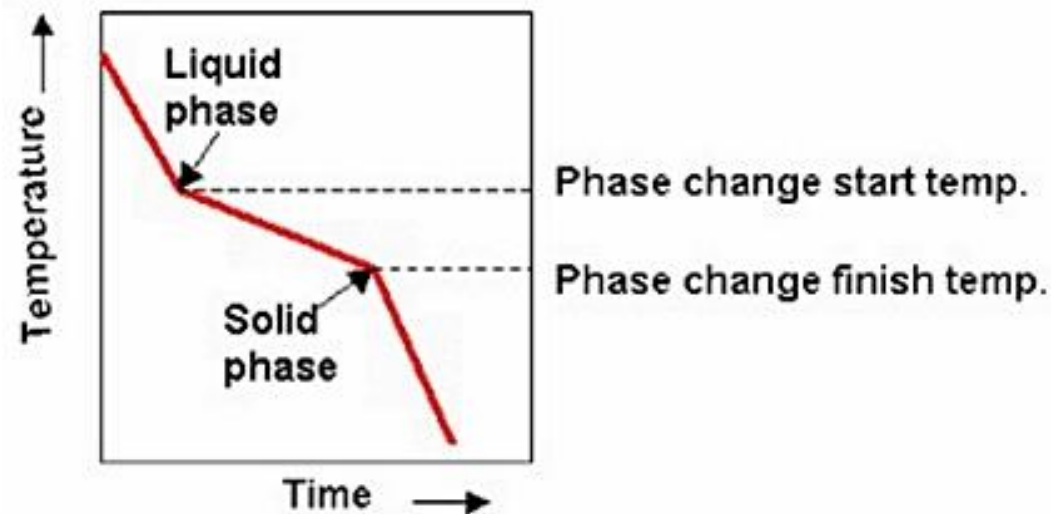
Cooling curve for a pure metal showing possible undercooling

- The transformation temperature, as shown on the equilibrium diagram, represents the point at which the free energy of the solid phase is equal to that of the liquid phase
- Thus, we may consider the transition, as given in a phase diagram, to occur when the free energy change (ΔG)



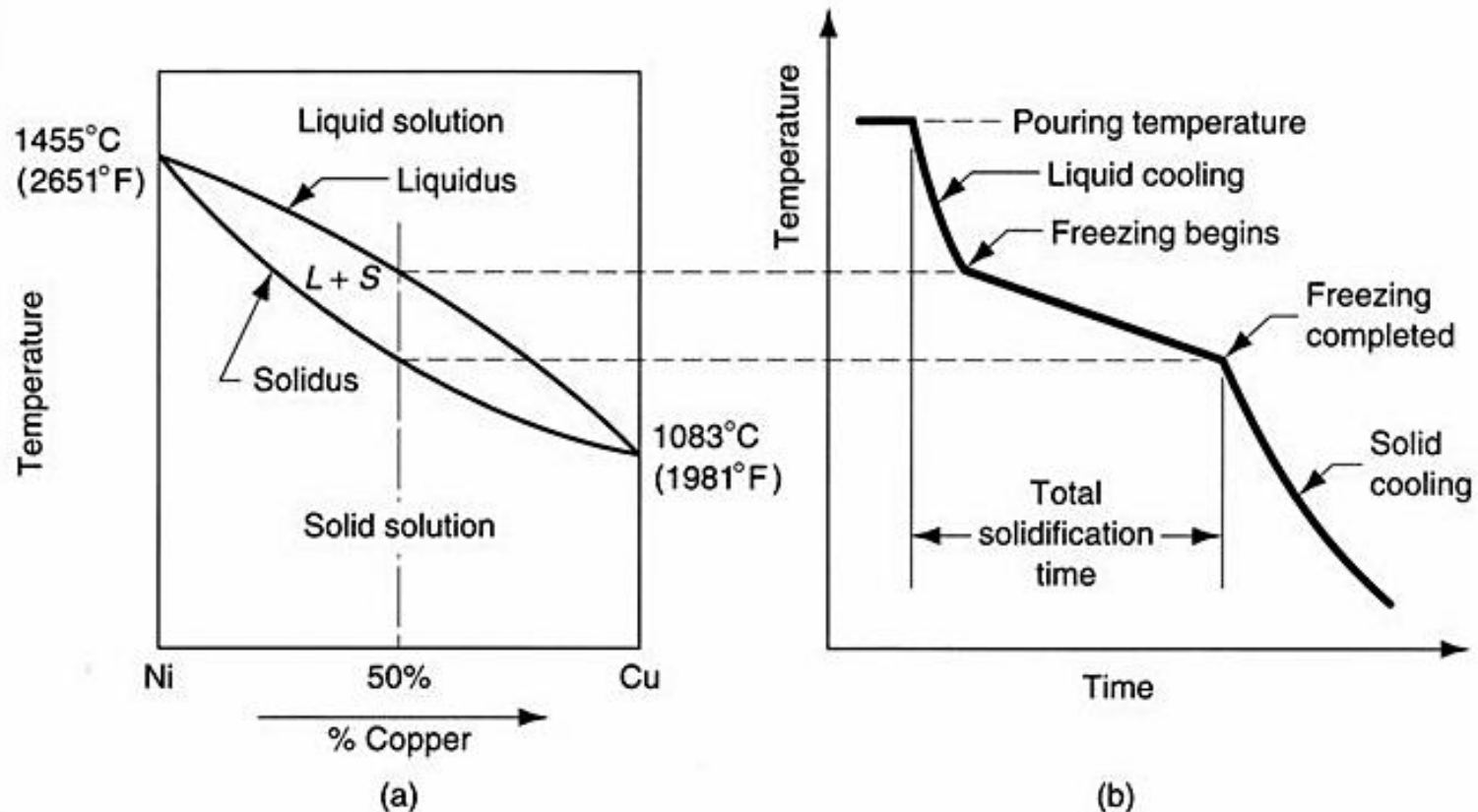
Cooling curve for a solid solution

- ❑ In case of alloys, the solidification does not take place at a constant temperature.
- ❑ In such cases, the solidification occurs in a range of temperature.



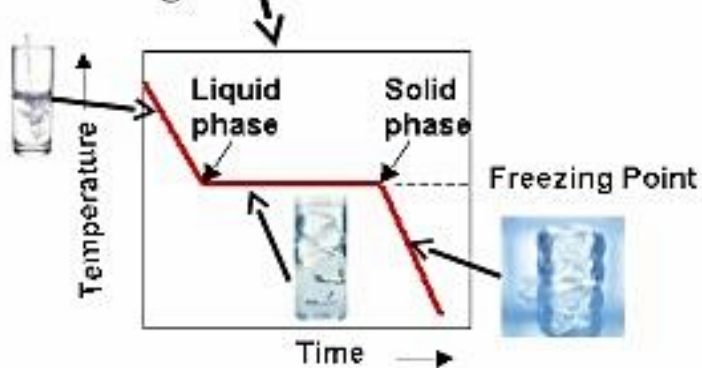
Cooling curve for a solid solution

- ❑ Solidification of alloys, most alloys freezing over a temperature range rather than a single temperature

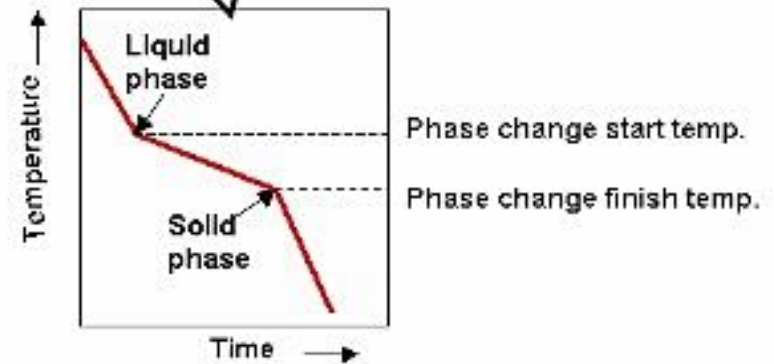


Comparing of Cooling Curve

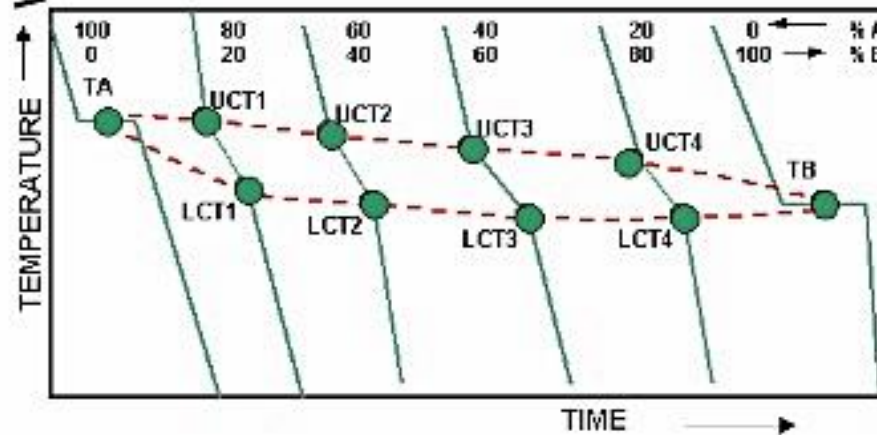
Cooling Curve of a Pure Metal.



Cooling Curve of a Solid Solution (An Alloy).



Set of Cooling Curves for various Solid Solutions.



LCT-Lower Critical Temperature

UCT-Upper Critical Temperature

TA-Melting Temperature of Alloy A

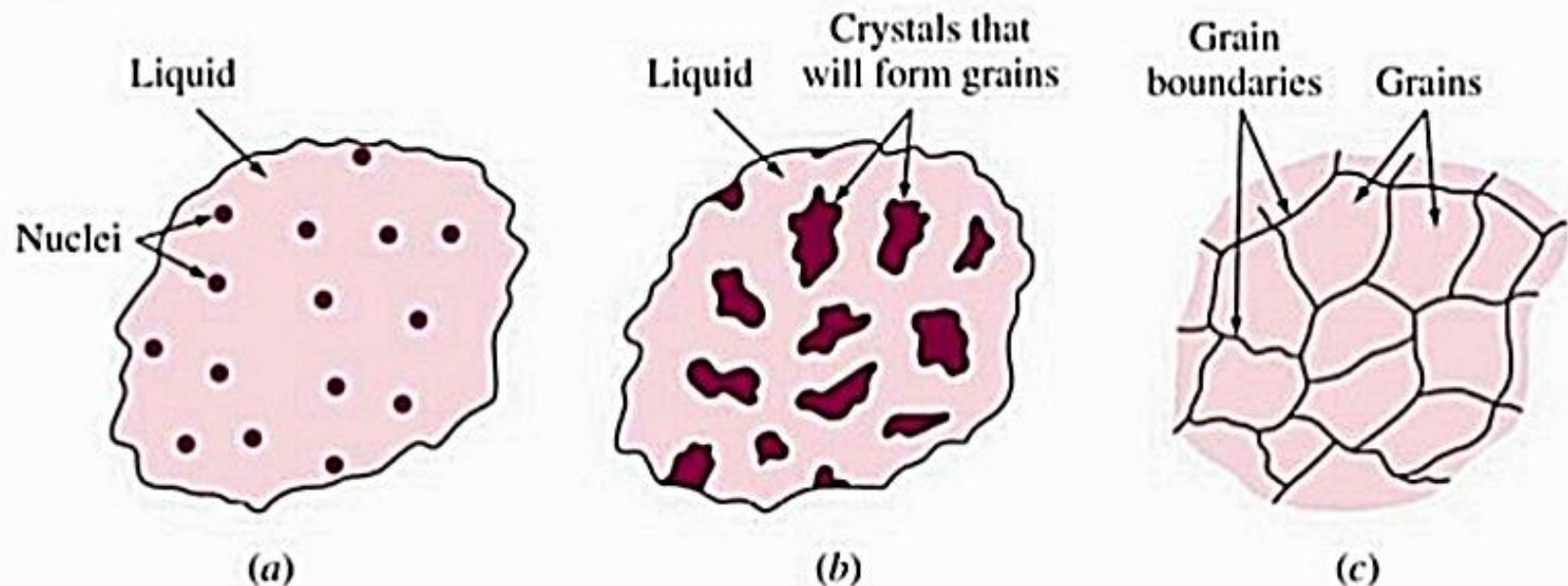
TB-Melting Temperature of Alloy B

Mechanism of solidification of metals

- ❑ The solidification of metals occur by **nucleation** and **growth** transformation.
- ❑ In nucleation and growth transformation, the nuclei of the solid phase are formed and then they grow.

Nucleation and growth transformation

- Nucleation – The physical process by which a new phase is produced in a material. In the case of solidification, this refers to the formation of tiny stable solid particles in the liquid.
- Growth - The physical process by which a new phase increases in size. In the case of solidification, this refers to the formation of a stable solid particle as the liquid freezes.



Nucleation and growth transformation

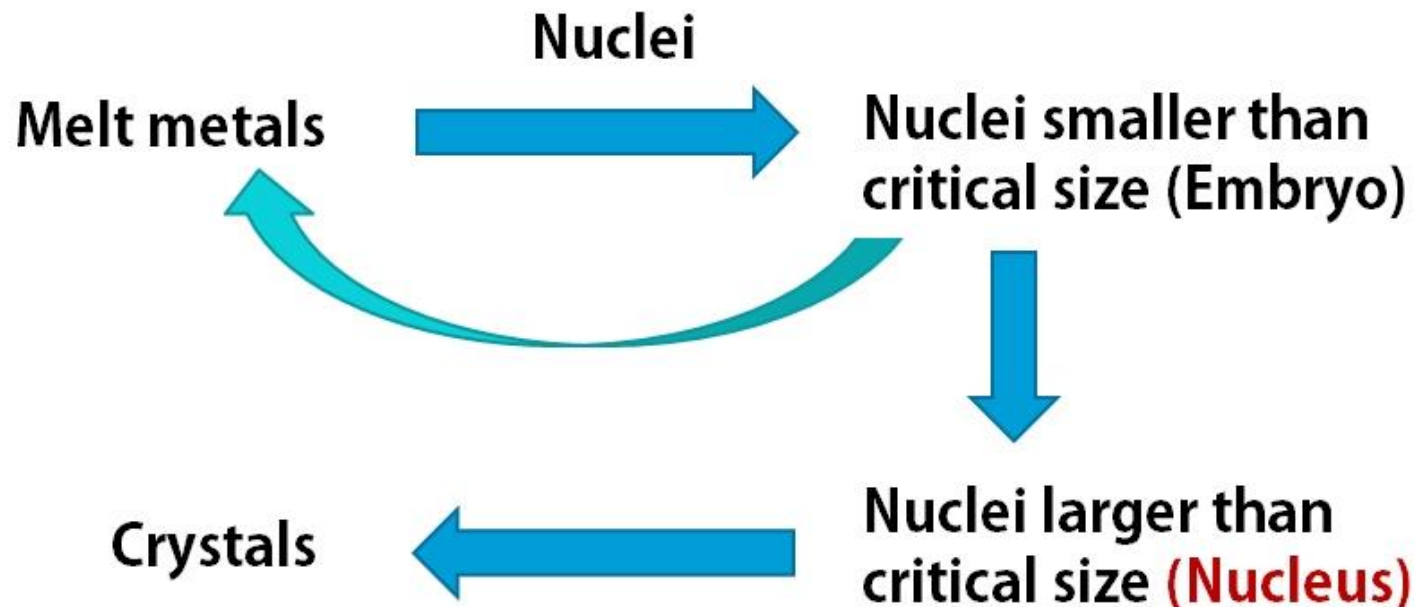
□ The nucleation and growth transformation may be of two types.

1. Homogeneous nucleation
2. Heterogeneous nucleation



Homogeneous nucleation

- Homogeneous nucleation—formation of a critically sized solid from the liquid by clustering together of a large number of atoms at a high undercooling (without an external interface)



Homogeneous nucleation

Energy of homogeneous nucleation with consist of 3 types of energy change:

1) Volume or bulk free energy, ΔG_v

- bulk free energy is started of metal solidification, with negative value (–)

$$\frac{4}{3}\pi r^3 \Delta G_v$$

ΔG_v is changing of free energy between liquid and solid per unit volume of metal

Homogeneous nucleation

2) Surface free energy, ΔG_s

- surface free energy is free energy between surface of nucleus and melt metal at surrounding of nucleus, with positive value (+)

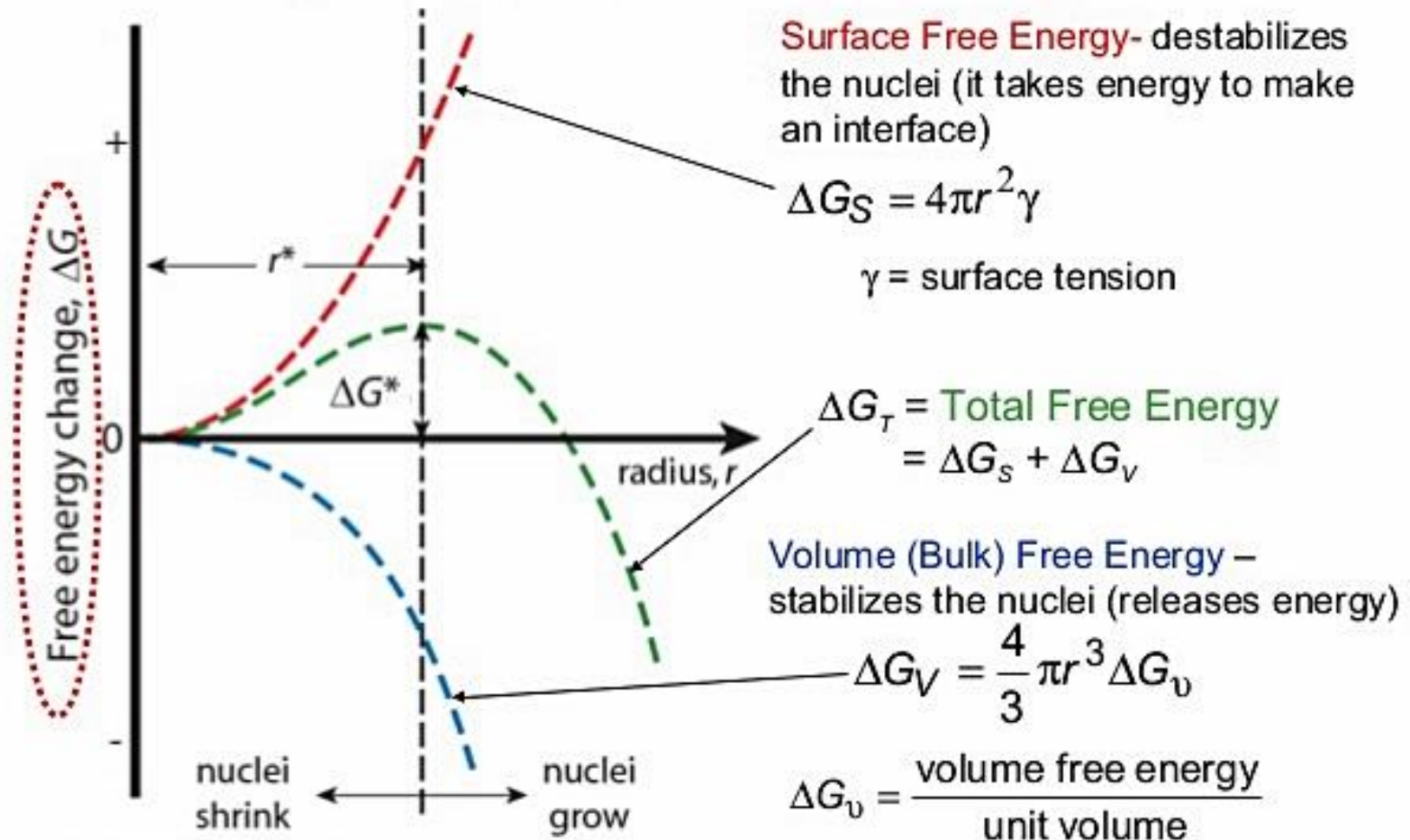
$$4\pi r^2\gamma$$

γ is surface tension

3) Total free energy, ΔG_T

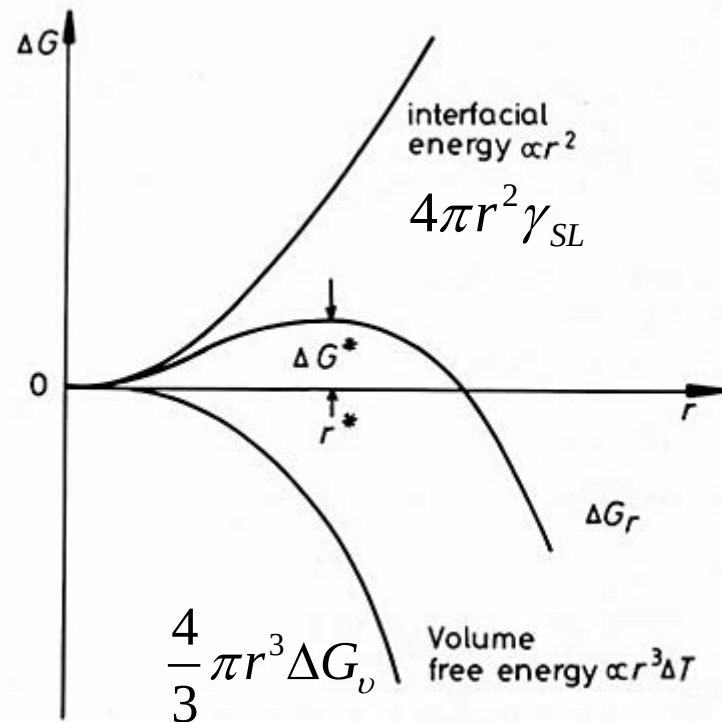
$$\begin{aligned}\Delta G_T &= \Delta G_v + \Delta G_s \\ \Delta G_T &= \frac{4}{3}\pi r^3 \Delta G_v + 4\pi r^2 \gamma\end{aligned}$$

Homogeneous nucleation & Energy effects



r^* = critical nucleus: for $r < r^*$ nuclei shrink; for $r > r^*$ nuclei grow (to reduce energy)

Homogeneous nucleation



•By assuming that solid phase nucleates are spherical ‘clusters’ of radius, r , one can define the excess of free energy change for a single nucleus, $\Delta G_{(r)}$:

$$\Delta G_{(r)} = \frac{4}{3}\pi r^3 \Delta G_v + 4\pi r^2 \gamma_{SL}$$

where γ_{SL} is the solid/liquid interfacial energy.

The critical radius r^* (defined as the radius at which $\Delta G_{(r)}$ is maximum) is given by:

$$r^* = \frac{2\gamma_{SL}T_m}{\Delta H_f \Delta T}$$

The associated energy barrier to homogeneous nucleation, ΔG^* is:

$$\Delta G^* = \frac{16\pi\gamma_{SL}^3 T_m^2}{\Delta H_f^2 \Delta T^2}$$

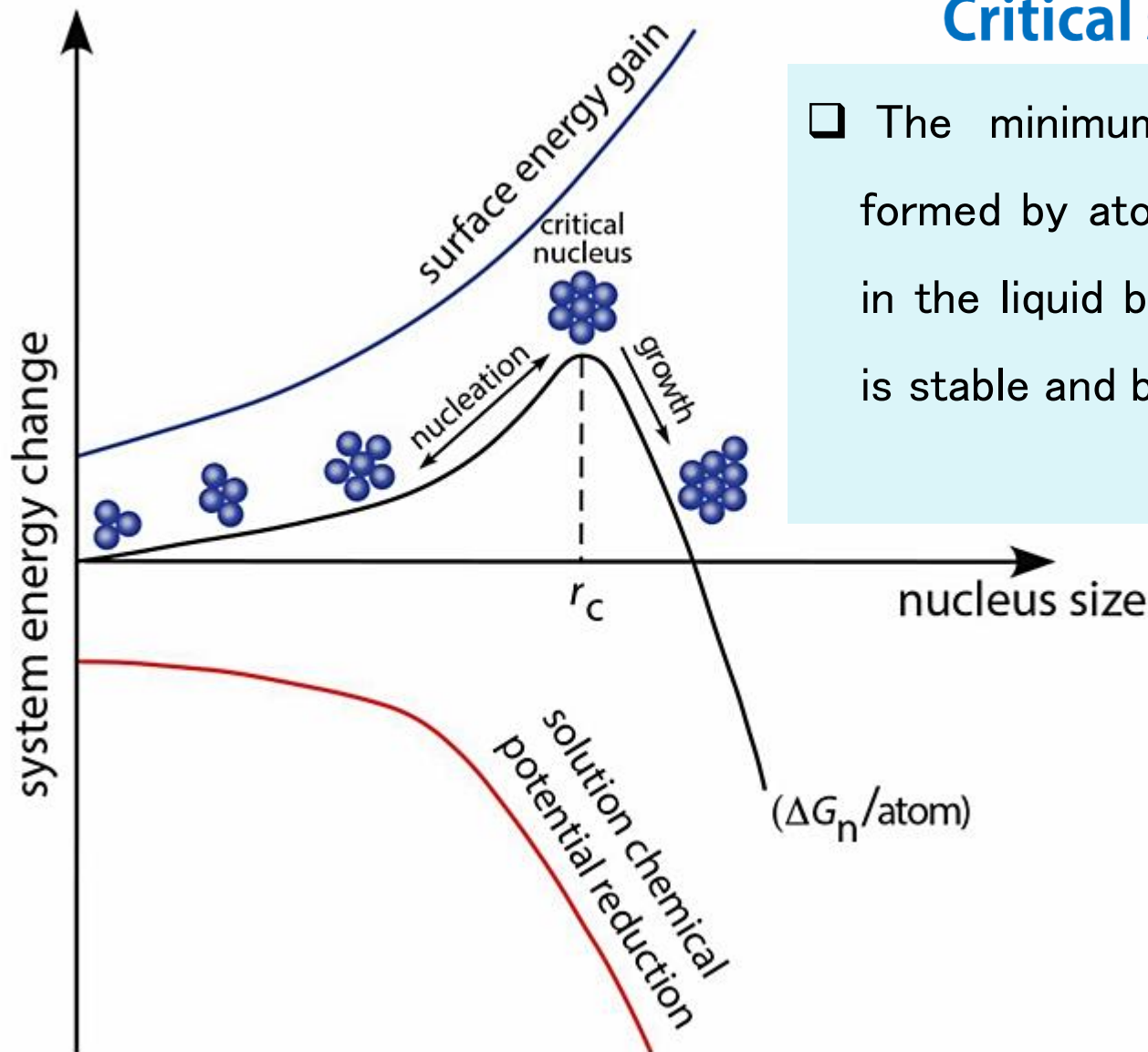
Important : the temperature-dependence of these terms are: $r^* \propto 1/\Delta T$ and $\Delta G^* \propto 1/\Delta T^2$

Higher under cooling – easy to form new phase !!

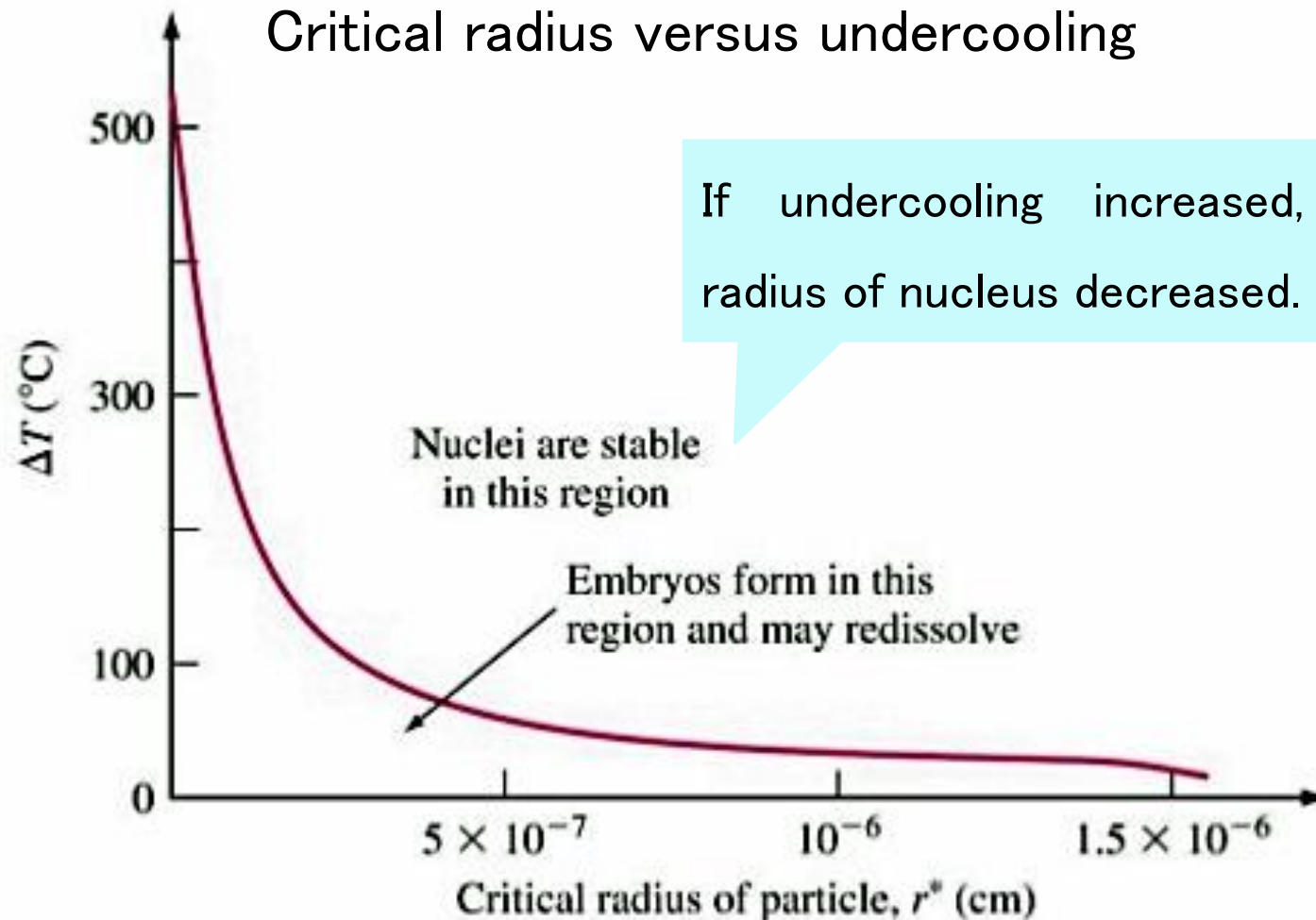
Homogeneous nucleation

Critical size of nucleus

- The minimum size that must be formed by atoms clustering together in the liquid before the solid particle is stable and begins to grow.



Homogeneous nucleation



Homogeneous nucleation

Calculation of Critical Radius for the Solidification

Calculate the size of the critical radius when solid **copper** forms by homogeneous nucleation.

Example 8.1 SOLUTION

$$\Delta T = 236 + 273 = 509K \qquad \Delta H_f = 1628J / cm^3$$

$$T_m = 1085 + 273 = 1385K \qquad \sigma_{SL} = 177 \times 10^{-7} J / cm^2$$

$$r^* = \frac{2\sigma_{SL}T_m}{\Delta H_f \Delta T} = \frac{(2)(1.77 \times 10^{-7} J / cm^2)(1385K)}{(1628J / cm^3)(509K)} = 5.80 \times 10^{-8} cm$$

TABLE 8-1 ■ *Values for freezing temperature, latent heat of fusion, surface energy, and maximum undercooling for selected materials*

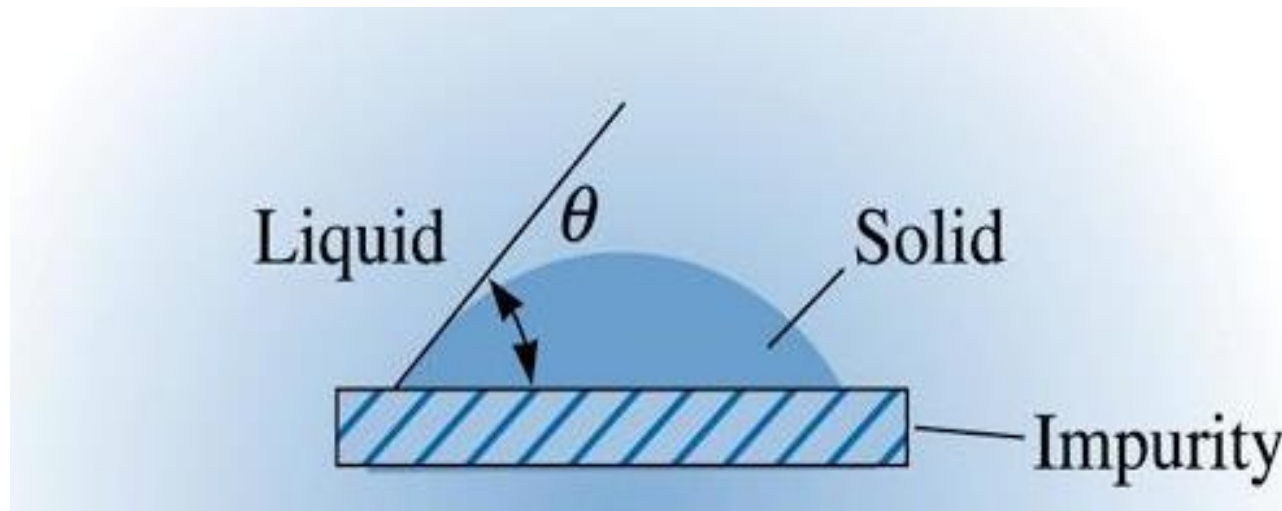
	Freezing Temperature (T_m)	Heat of Fusion (ΔH_f)	Solid-Liquid Interfacial Energy (σ_{sl})	Typical Undercooling for Homogeneous Nucleation (ΔT)
Metal	(°C)	(J/cm ³)	(J/cm ²)	(°C)
Ga	30	488	56×10^{-7}	76
Bi	271	543	54×10^{-7}	90
Pb	327	237	33×10^{-7}	80
Ag	962	965	126×10^{-7}	250
Cu	1085	1628	177×10^{-7}	236
Ni	1453	2756	255×10^{-7}	480
Fe	1538	1737	204×10^{-7}	420
NaCl	801			169
CsCl	645			152
H ₂ O	0			40

Heterogeneous nucleation

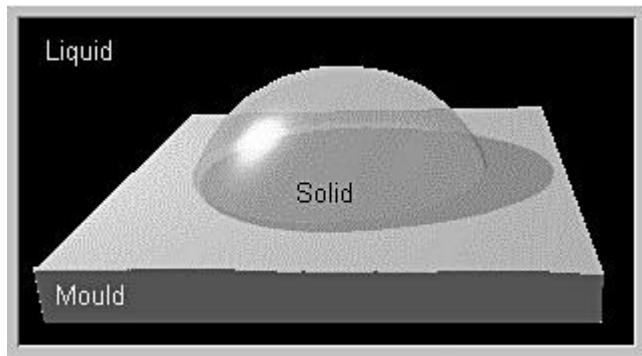
- ❑ Heterogeneous nucleation – formation of a critically sized solid from the liquid on an impurity surface.
- ❑ Heterogeneous nucleation occurs in a liquid on the surface of its container, insoluble impurities and other structural materials that lower the critical free energy required to form a stable nucleus.

Heterogeneous nucleation

- ❑ A solid forming on an impurity can be assumed to have a critical radius with a smaller increase in the surface energy.
- ❑ Thus, heterogeneous nucleation can occur with relatively low undercooling.



Heterogeneous nucleation



- Let us consider that a solid cluster forms on a mould wall. In this case, one can define the excess of **free energy** change for a nucleus, ΔG_r as follows:

$$\Delta G_{(r)} = \frac{1}{3} \times [A_{SL}\gamma_{SL} + A_{SM}\gamma_{SM}]$$

- It can be shown that the critical radius of heterogeneous nucleation is **equal** to r_{hom}^*

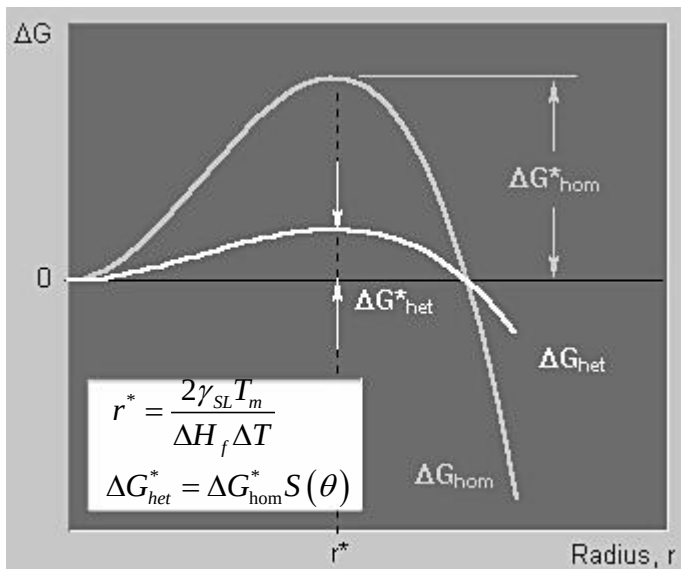
$$r^* = \frac{2\gamma_{SL}T_m}{\Delta H_f \Delta T} = r_{hom}^*$$

- And:

$$\Delta G_{het}^* = \frac{16\pi\gamma_{SL}^3 T_m^2}{\Delta H_f^2 \Delta T^2} S(\theta) = \Delta G_{hom}^* S(\theta)$$

- When $S(\theta)$ is a geometrical factor and θ is a wetting angle:

$$S(\theta) = \frac{(2 + \cos \theta)(1 - \cos \theta)^2}{4}$$



Note: Heterogeneous nucleation is faster than Homogeneous process

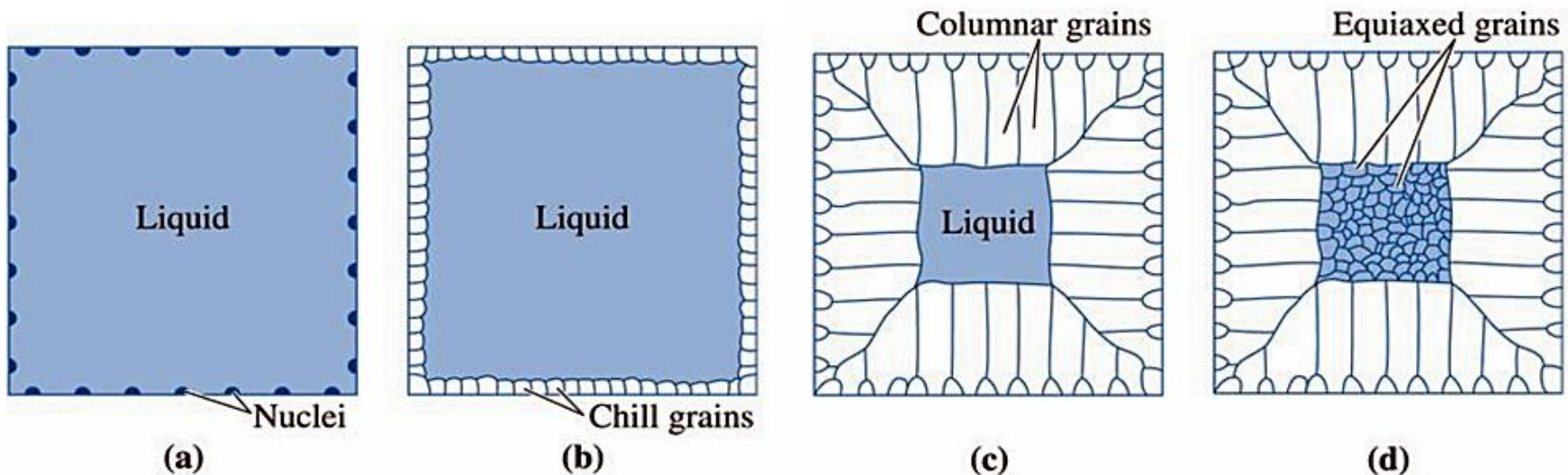
Solidification of casting

Casting Structure

- **Chill zone** – A region of small, randomly oriented grains that forms at the surface of a casting as a result of heterogeneous nucleation.
- **Columnar zone** – A region of elongated grains having a preferred orientation that forms as a result of competitive growth during the solidification of a casting.
- **Equiaxed zone** – A region of randomly oriented grains in the center of a casting produced as a result of widespread nucleation.

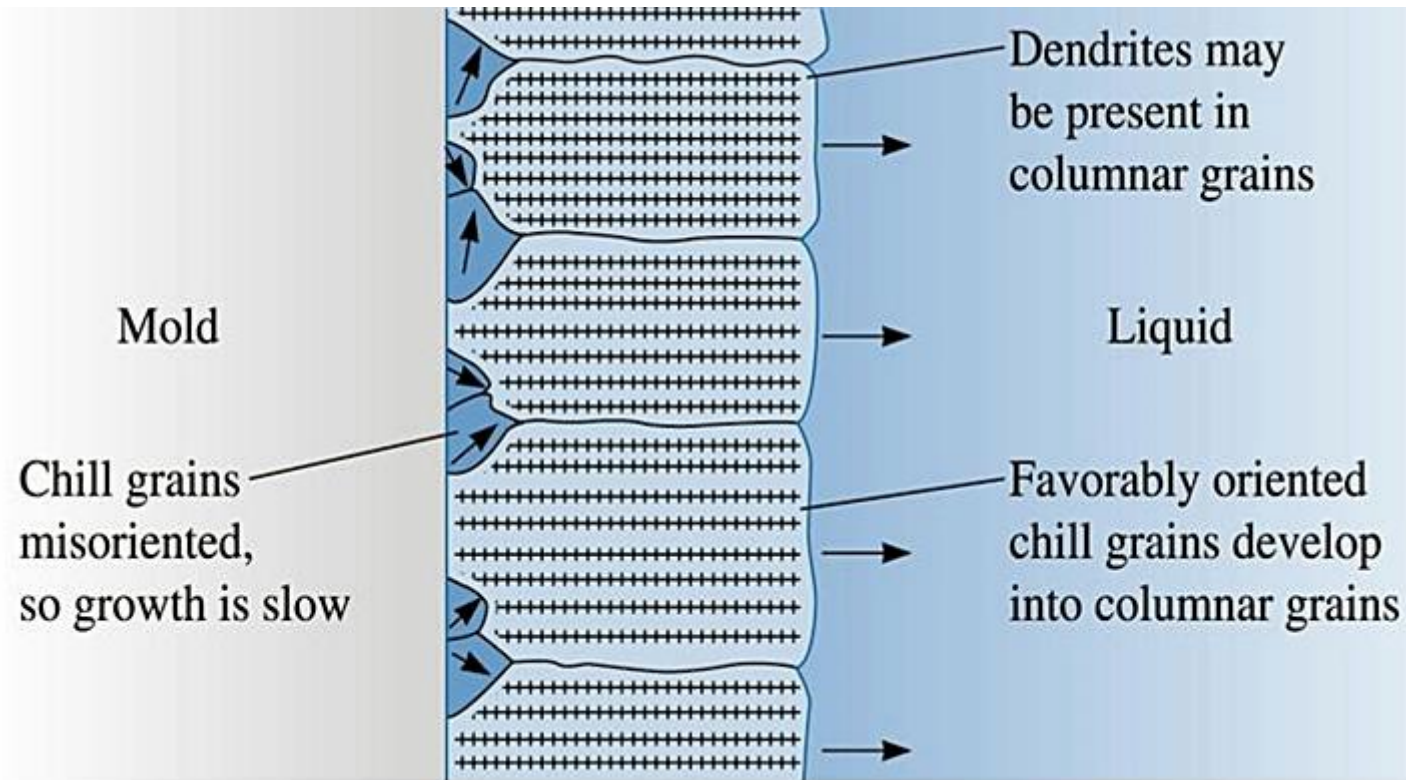
Solidification of casting

□ Development of the ingot structure of a casting during solidification:



(a) Nucleation begins, (b) the chill zone forms, (c) preferred growth produces the columnar zone, and (d) additional nucleation creates the equiaxed zone

Solidification of casting



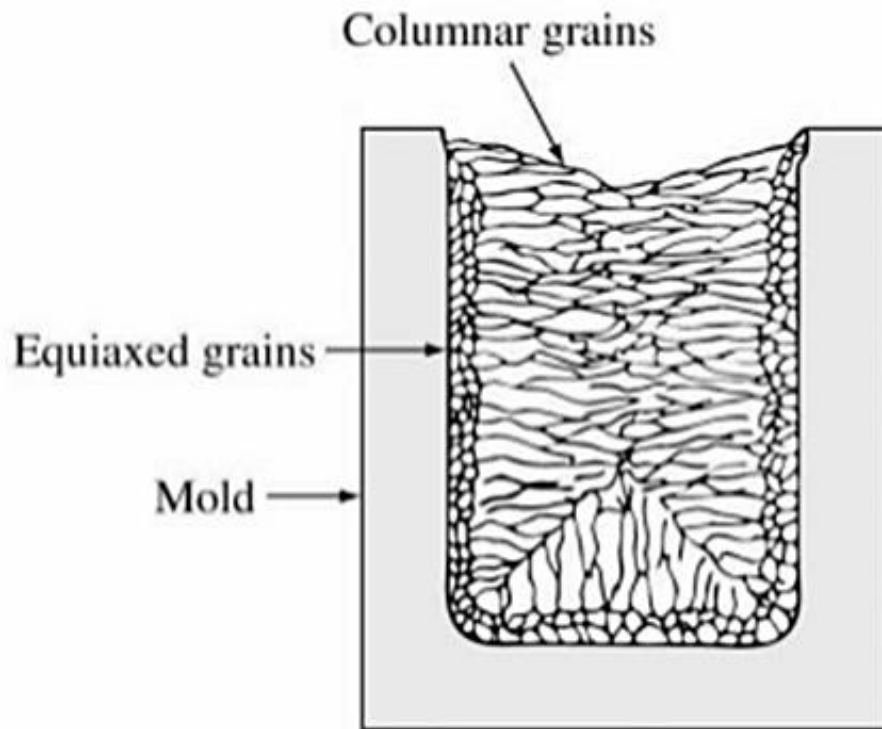
Competitive growth of the grains in the chill zone results in only those grains with favorable orientations developing into columnar grains

Solidification of casting

❑ Crystal growth and grain formation

- Nuclei → Crystals → Grains
- Polycrystalline: solidified metal containing many crystals.
- Grain: crystals in solidified metal.
- Grain boundaries: the surfaces between the grains.
- Two major types of grain structure:
 1. **Equiaxed grains**— crystals grow about equally in all direction, commonly found adjacent to a cold mould wall.
 2. **Columnar grains**— long, thin, coarse grains, created when metal solidifies rather slow in the presence of a steep temperature gradient. Columnar grains grow perpendicular to the mould surface

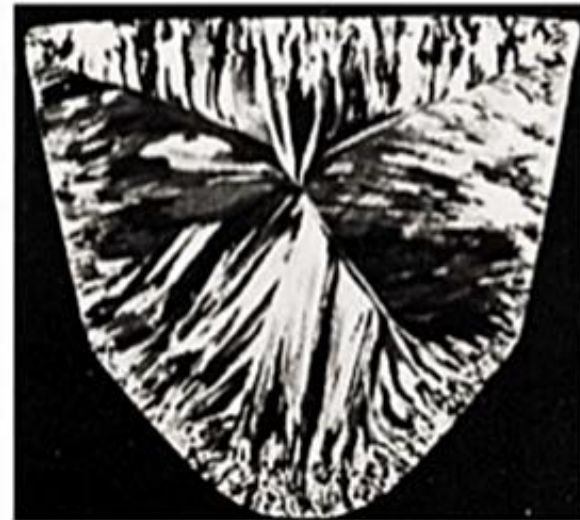
Solidification of casting



(a)

Side view

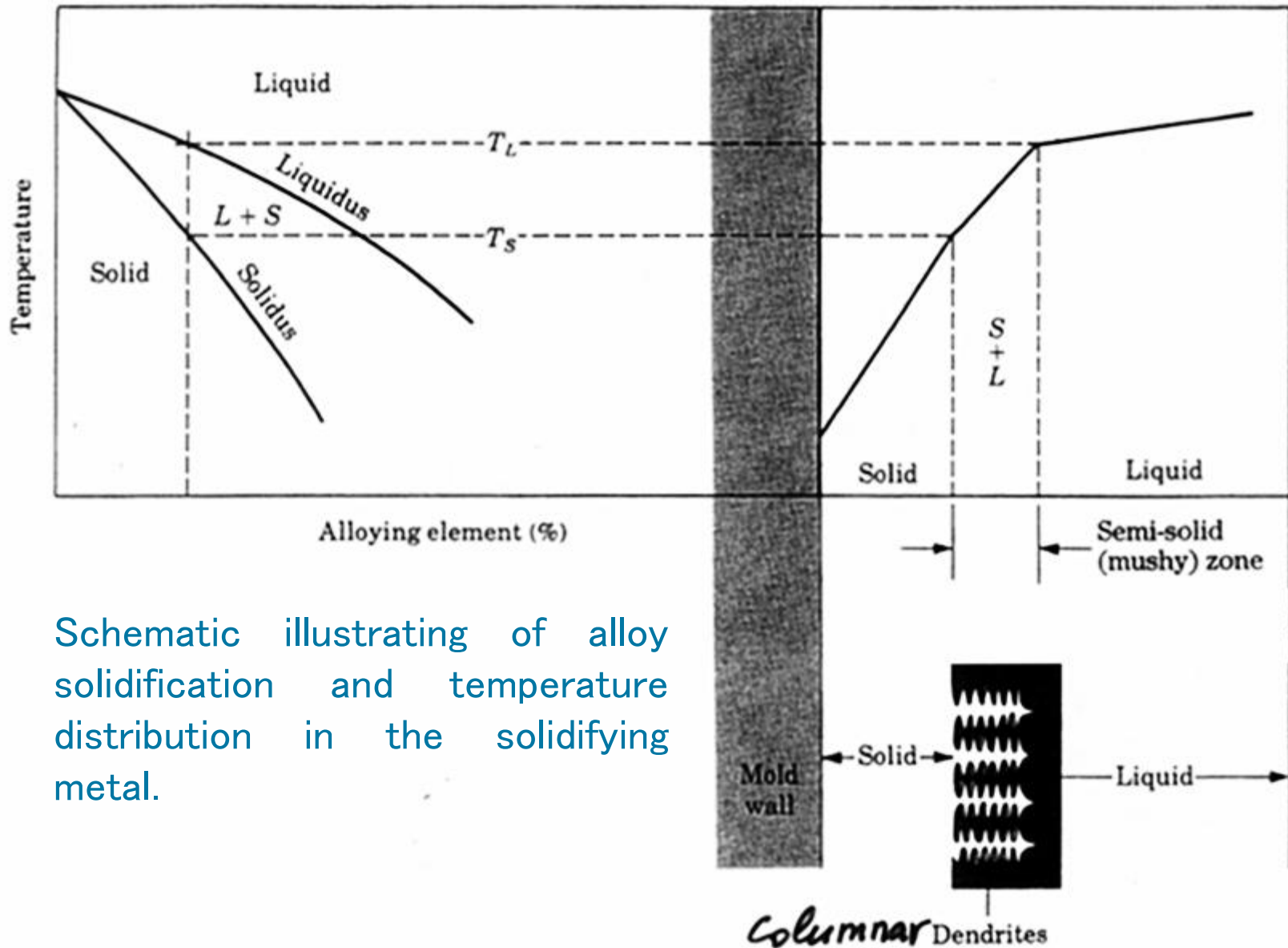
Al ingot



(b)

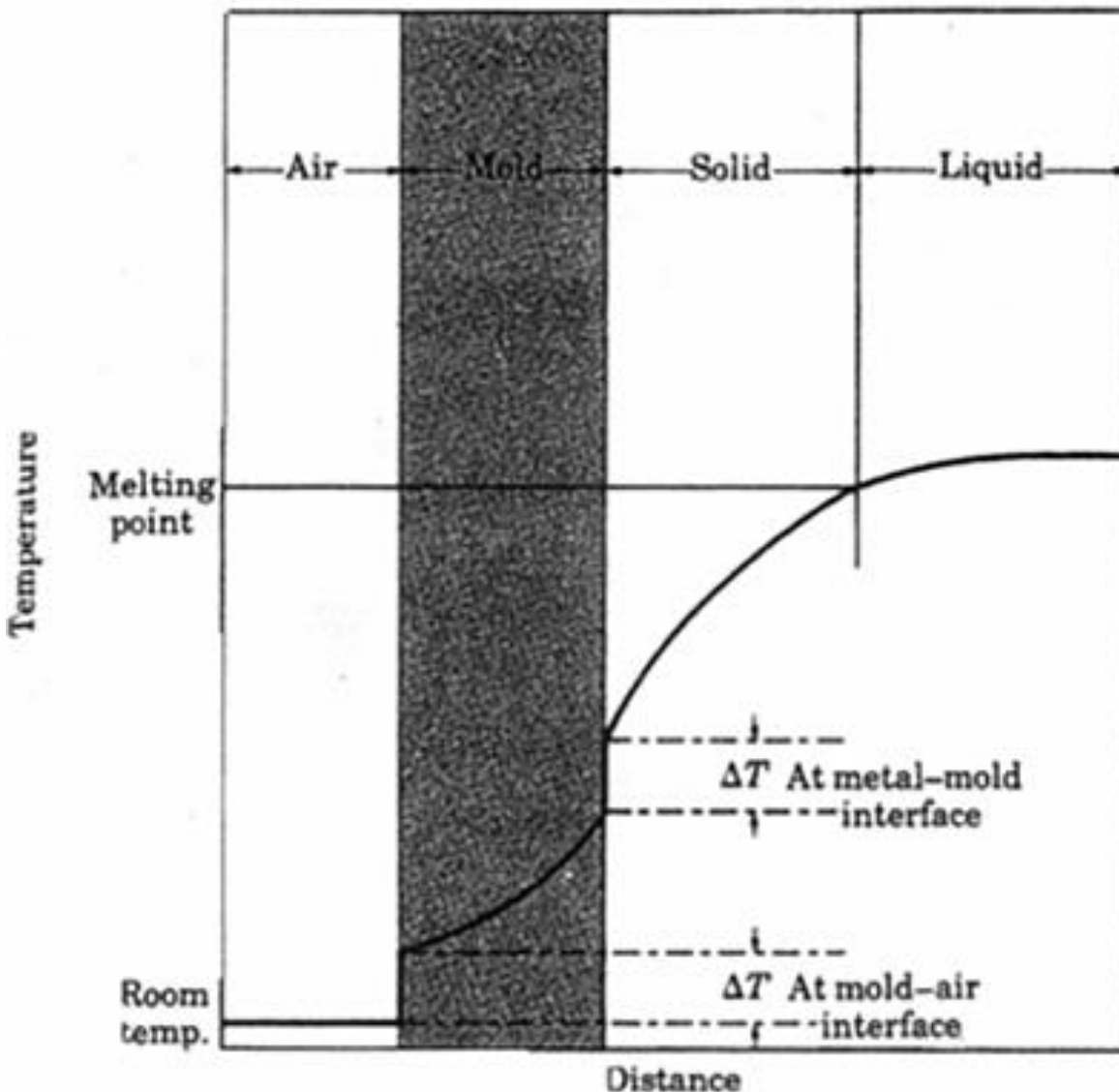
Top view

Solidification of casting



Schematic illustrating of alloy solidification and temperature distribution in the solidifying metal.

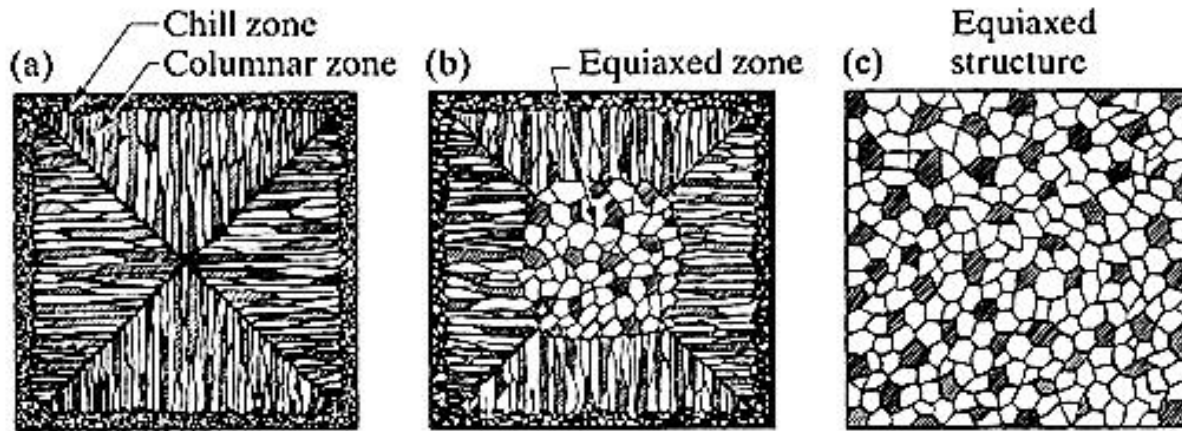
Solidification of casting



Heat transfer:

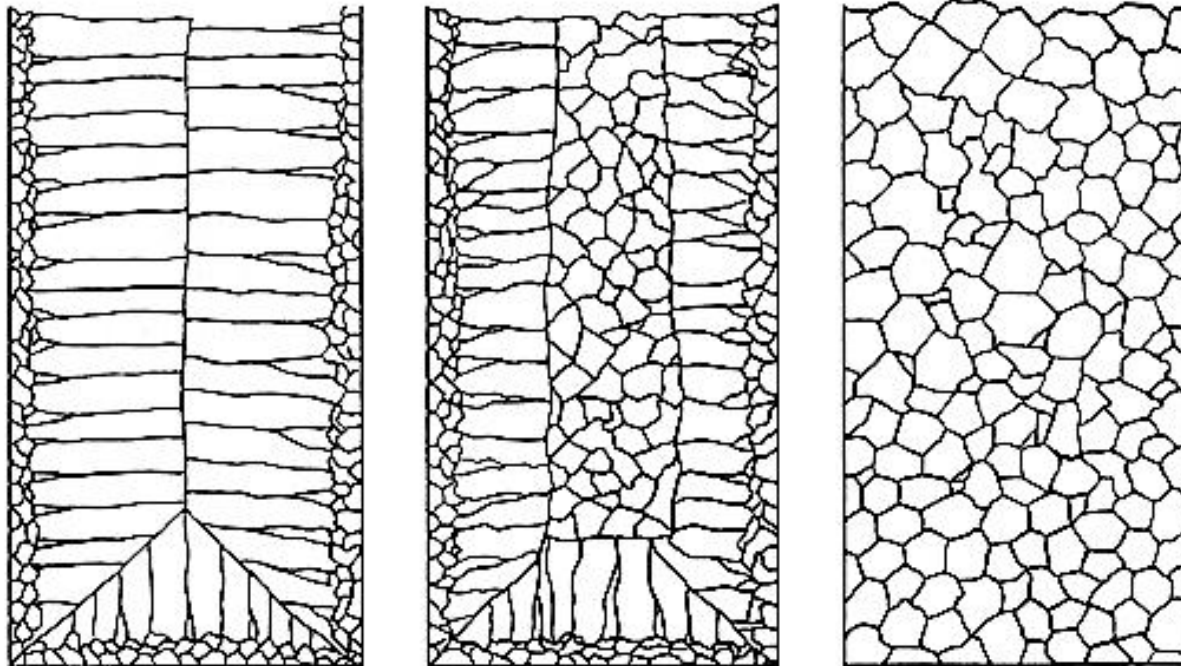
- from pouring to solidification and cooling to room temperature.
- it depends on many factors related to the casting material and the mold and process parameters.

Solidification of casting



Three cast structure of metals solidified in a square mold:

- (a) Pure metals
- (b) Solid-solution alloys
- (c) Structure obtained by using nucleating agents

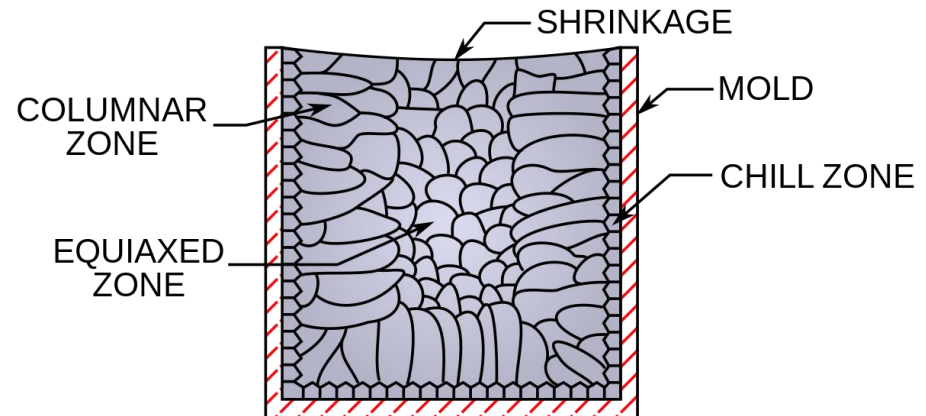


Source: G.W. Form, J.F. Wallace, J.L. Walker, and A. Cibula

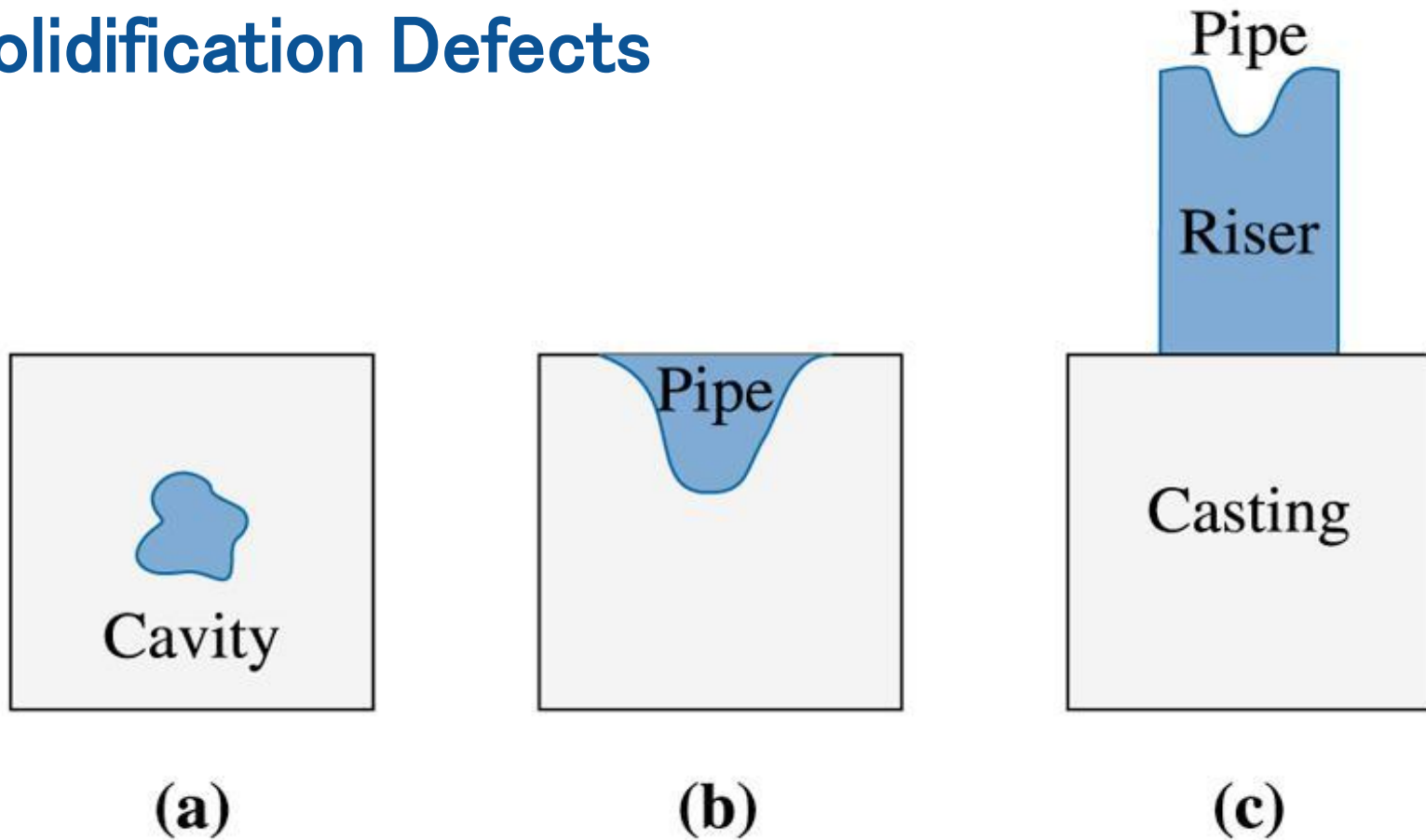
Solidification Defects

□ **Shrinkage:** contraction of a casting during solidification and cooling.

- Contraction of the liquid as it cools prior to its solidification
- Contraction during phase change from a liquid to solid
- Contraction of the solid as it continues to cool to ambient temperature

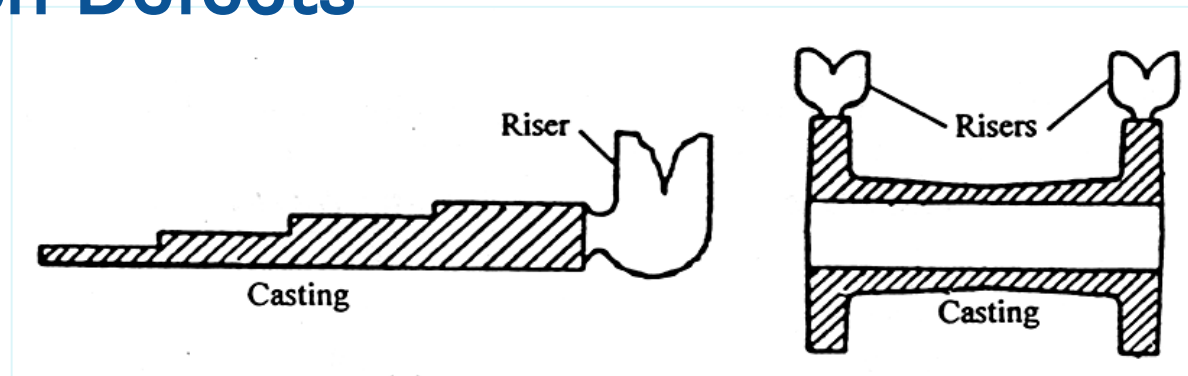


Solidification Defects

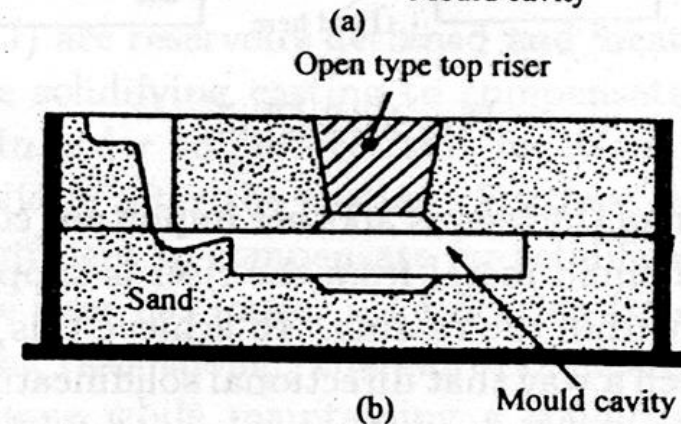
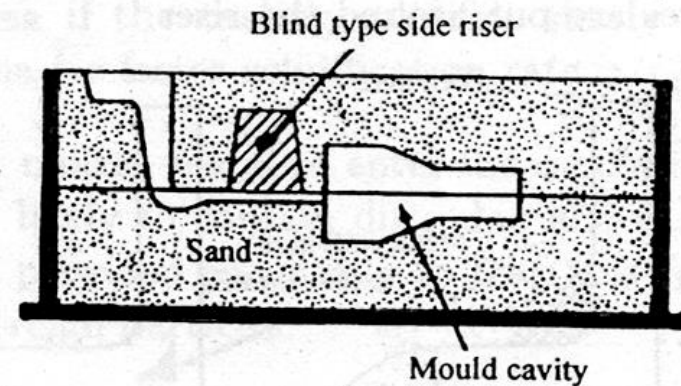


Several types of shrinkage can occur, including cavities and pipes. Risers can be used to help compensate for shrinkage

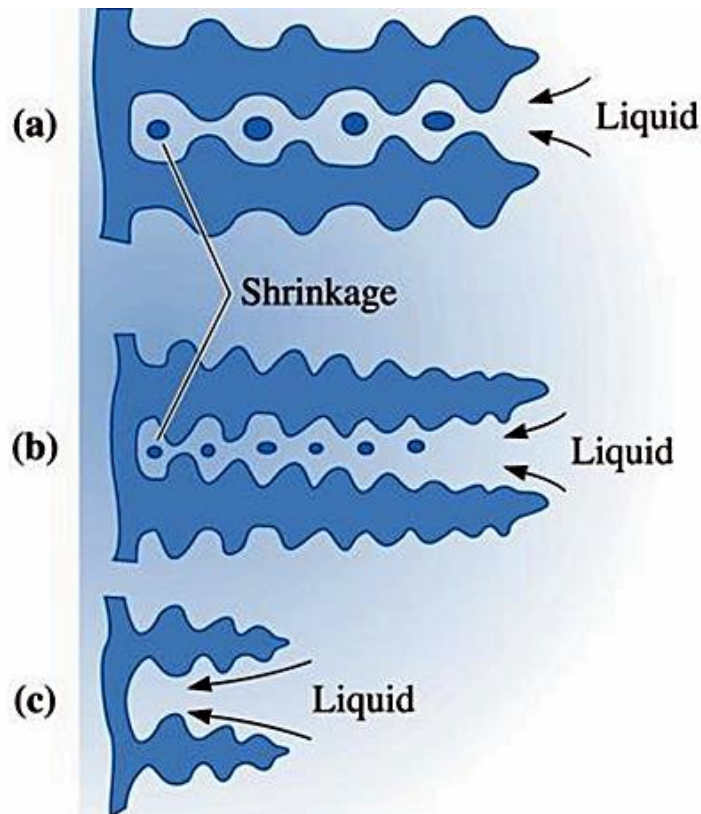
Solidification Defects



Location of Risers and Open and Closed Risers



Solidification Defects



Interdentritic shrinkage in an aluminum alloy is shown (x 80)



- (a) Shrinkage can occur between the dendrite arms.
- (b) Small secondary dendrite arm spacing result in smaller, more evenly distributed shrinkage porosity.
- (c) Short primary arms can help avoid shrinkage.

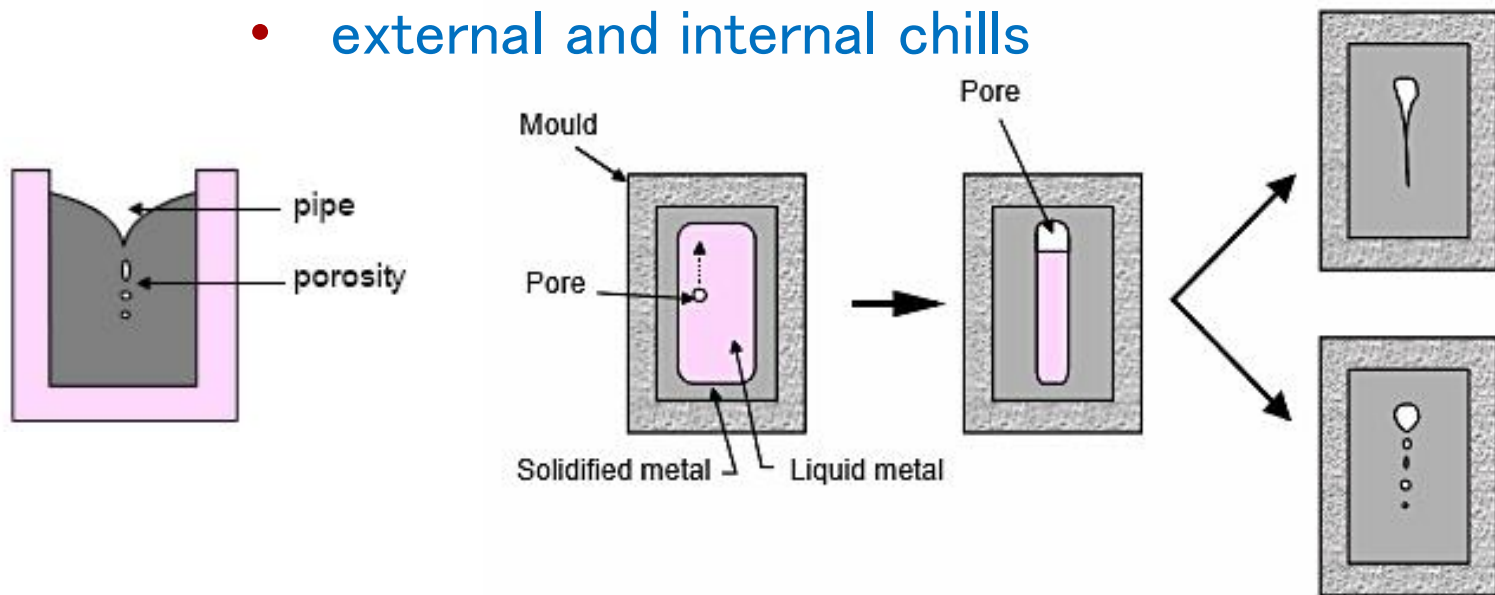
Solidification Defects

TABLE 8-2 ■ *Shrinkage during solidification for selected materials*

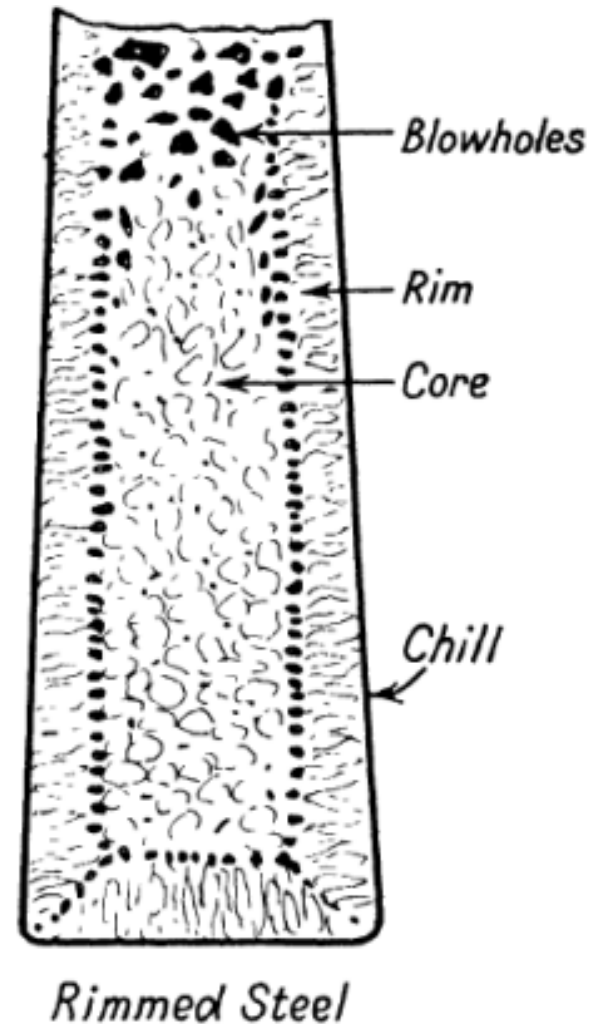
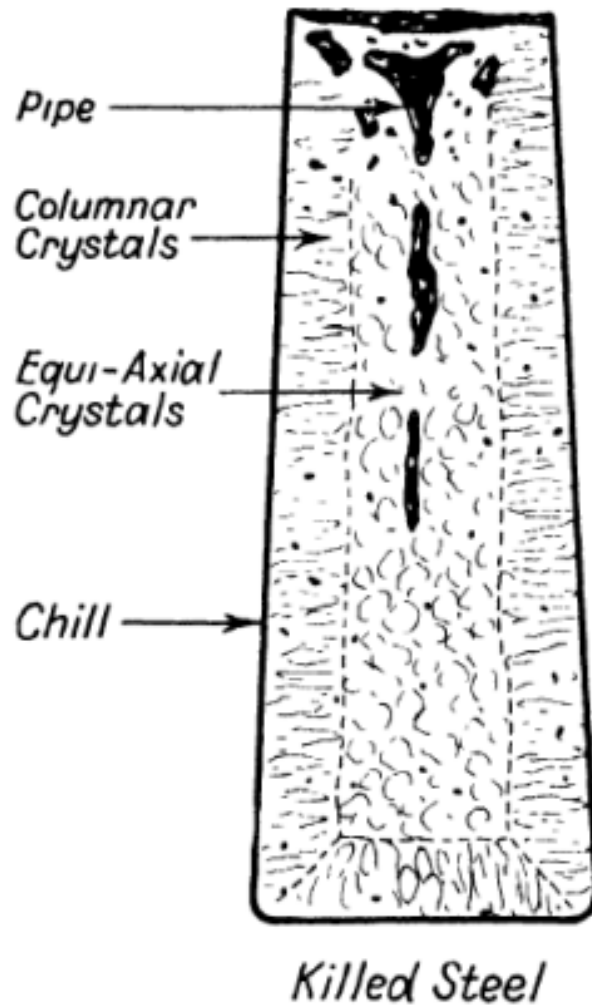
Material	Shrinkage (%)
Al	7.0
Cu	5.1
Mg	4.0
Zn	3.7
Fe	3.4
Pb	2.7
Ga	+3.2 (expansion)
H ₂ O	+8.3 (expansion)
Low-carbon steel	2.5–3.0
High-carbon steel	4.0
White Cast Iron	4.0–5.5
Gray Cast Iron	+1.9 (expansion)

Solidification Defects

- **Porosity:** caused by shrinkage or trapped gases, or both
 - porosity is detrimental to the ductility of a casting and its surface finish
 - porosity caused by shrinkage can be reduced or eliminated by various means:
 - adequate liquid metal feeding
 - external and internal chills

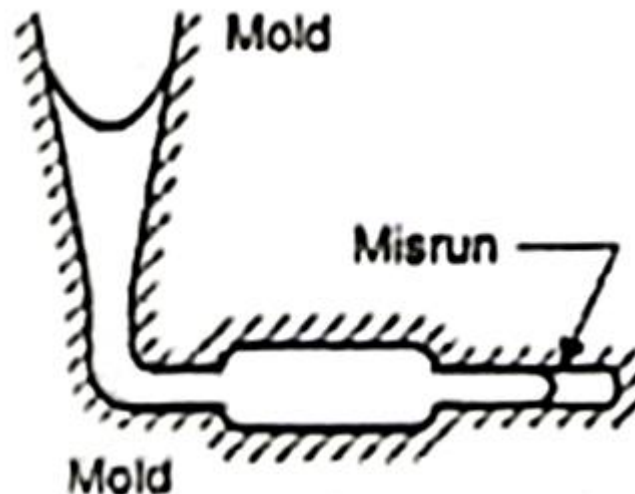


Solidification Defects



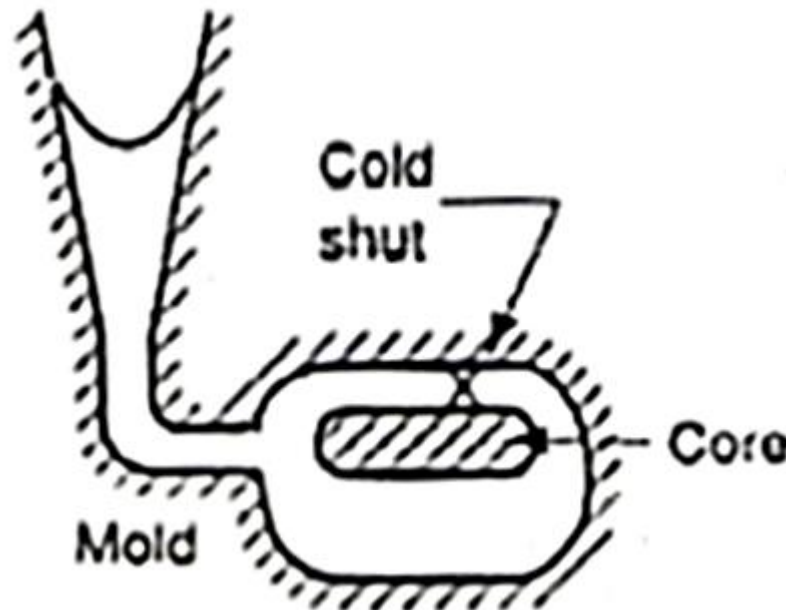
Solidification Defects

- ❑ **Misruns** : a misruns is a casting that has solidified before completely filling the mold cavity. Typical causes include
- 1) Fluidity of the molten metal is insufficient
 - 2) Pouring Temperature is too low
 - 3) Pouring is done too slowly and/or
 - 4) Cross section of the mold cavity is too thin.



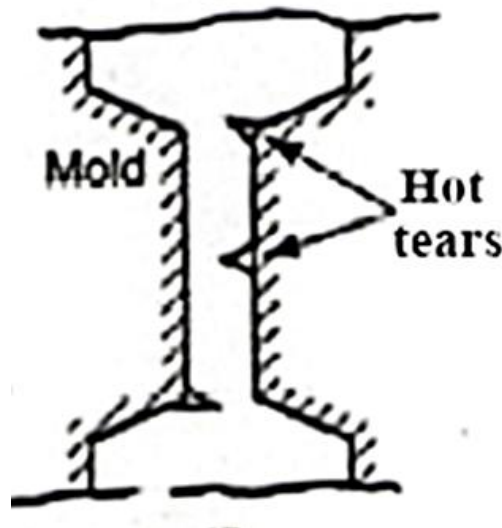
Solidification Defects

- ❑ **Cold Shut:** a cold shut occurs when two portion of the metal flow together, but there is lack of fusion between them due to premature freezing, Its causes are similar to those of a misruns.



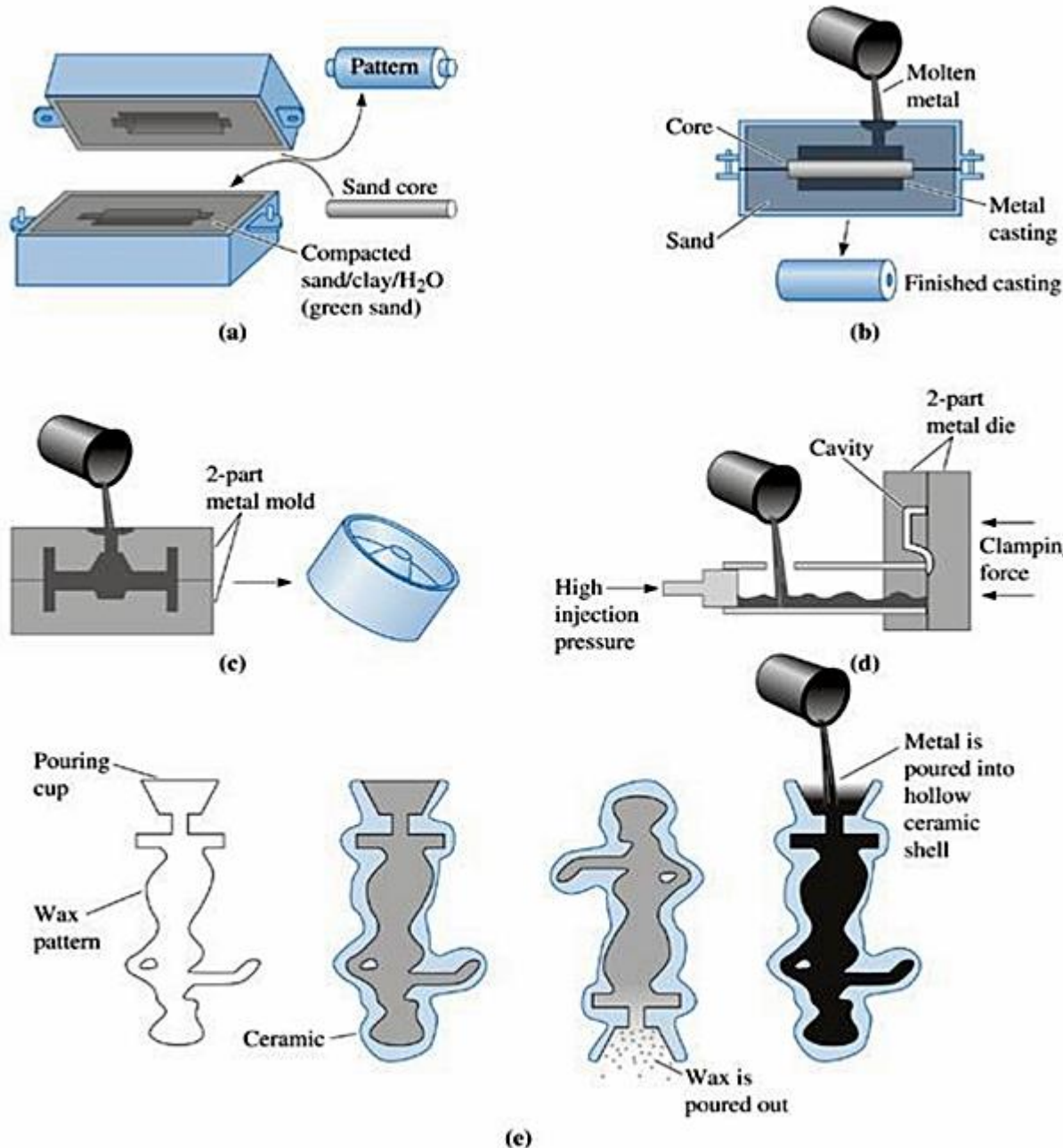
Solidification Defects

- ❑ **Hot Tearing:** this defect, also called hot cracking, occurs when the casting is restrained or early stages of cooling after solidification. The defect is manifested as a separation of the metal (hence the terms tearing or cracking) at a point of high tensile stress caused by metal's inability to shrink naturally.



Casting Processes for Manufacturing Components

- ❑ **Sand casting**: a casting process using sand molds.
- ❑ **Investment casting**: a casting process that is used for making complex shapes such as turbine blades, also known as the lost wax process.
- ❑ **Lost foam process**: a process in which a polymer foam is used as a pattern to produce a casting.
- ❑ **Permanent mold casting**: a casting process in which a mold can be used many times.
- ❑ **Pressure die casting**: a casting process in which molten metal/alloys is forced into a die under pressure.



Four typical casting processes: (a) and (b) Sand molding, in which clay-bonded sand is packed around a pattern.

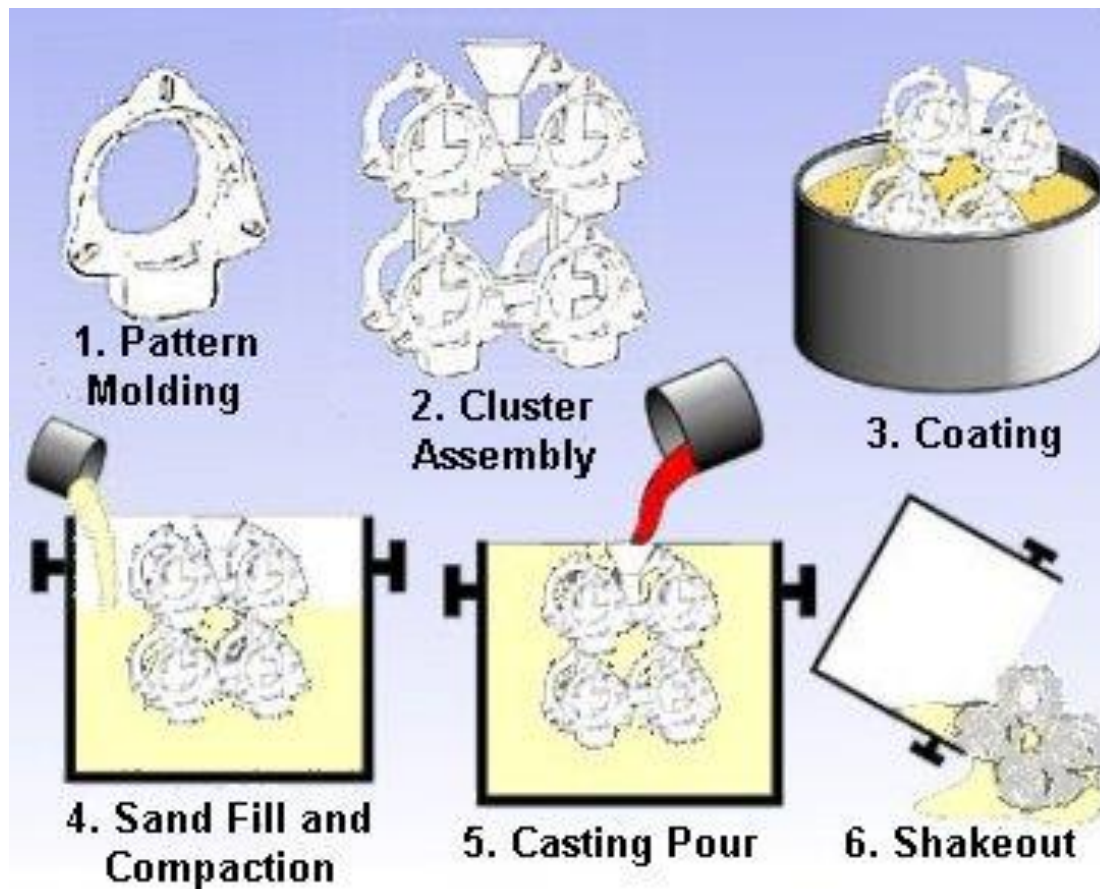
(c) The permanent mold process, in which metal is poured into an iron or steel mold.

(d) Die casting, in which metal is injected at high pressure into a steel die.

(e) Investment casting, in which a wax pattern is surrounded by a ceramic; after the wax is melted and drained, metal is poured into the mold.

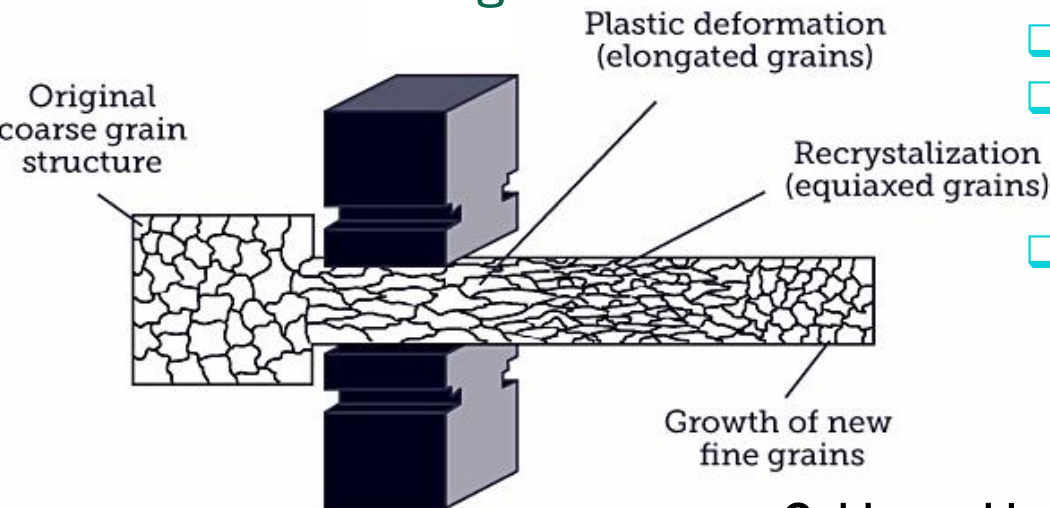
Casting Processes for Manufacturing Components

❑ Lost foam casting process



Recovery, Recrystallization and Grain Growth

❑ Cold-working

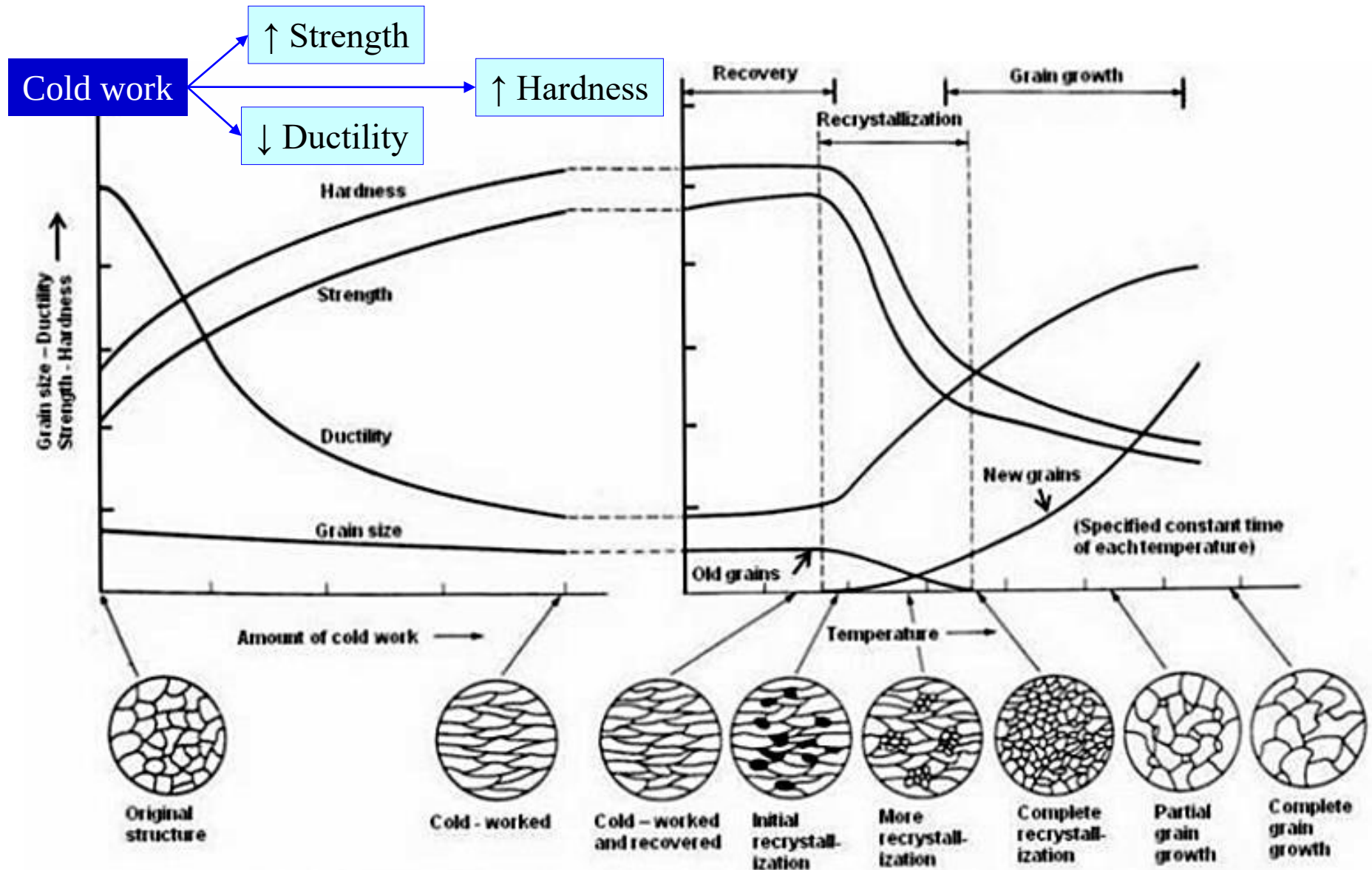


- ❑ Cold worked metals become **brittle**.
- ❑ Reheating, which increases ductility results in recovery, recrystallization and **grain growth**.
- ❑ This is called **annealing** and changes material properties.

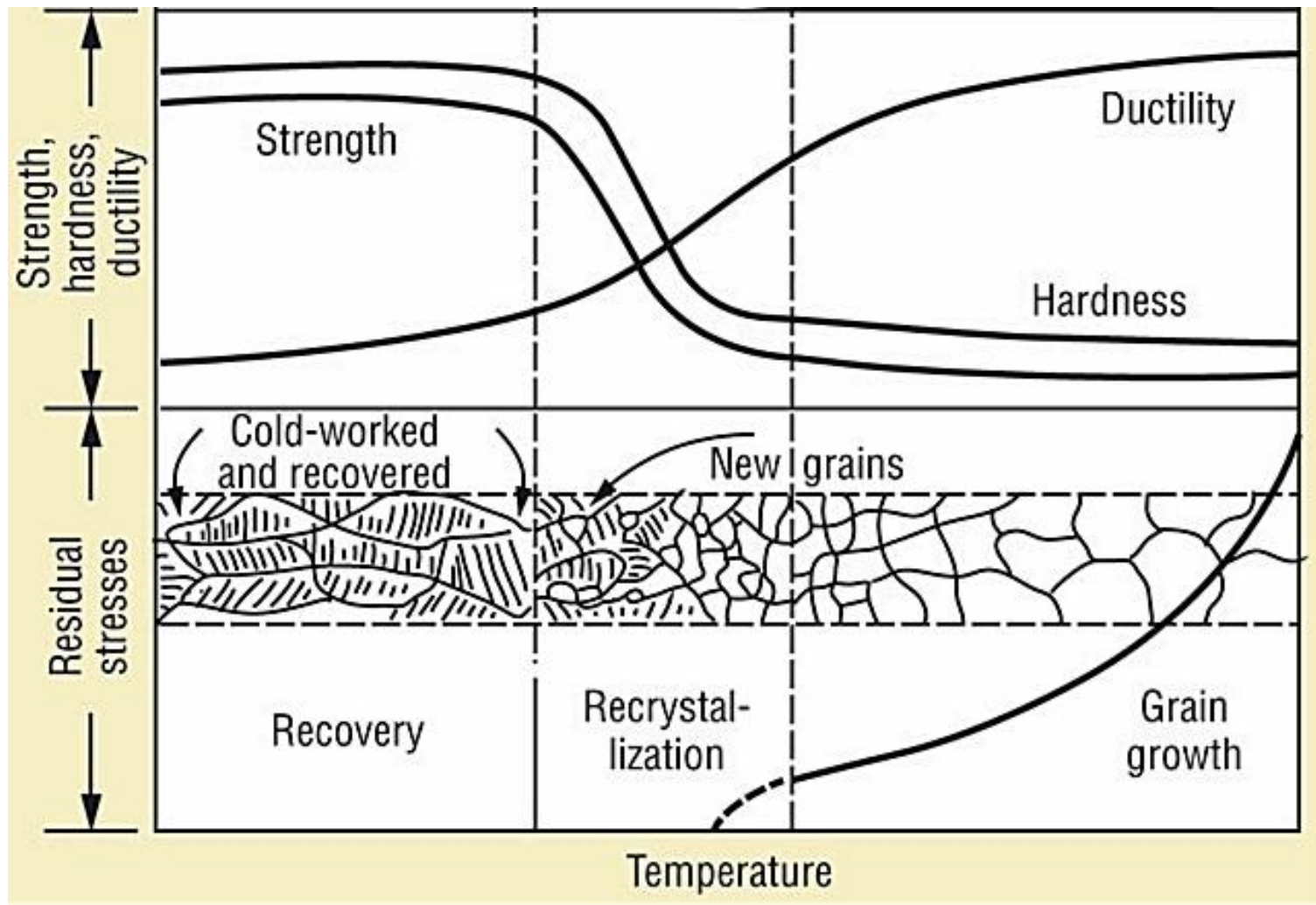
- Cold working is occurred of **plastic deformation** (**changes grain size** and **elongated grains** in the rolling direction) (**Recovery**).

- **Annealing** at $0.5T_m$, created new grains interstitial between original grains with rapid growth of new grains until lose of original grains (**Recrystallization**) after that new grains are slowly growth and stop growth (**Grain growth**).

Recovery, Recrystallization and Grain Growth



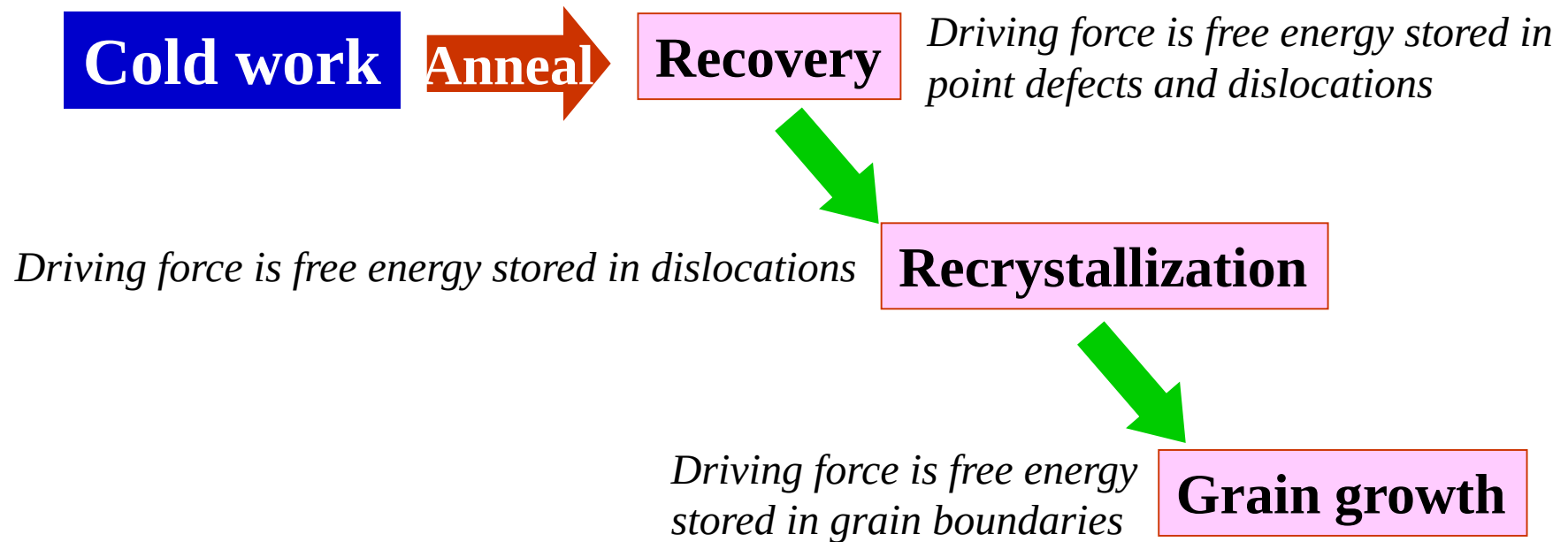
Recovery, Recrystallization and Grain Growth



Recovery, Recrystallization and Grain Growth

- ❑ **Recovery**: start nucleation of new grains.
- ❑ **Recrystallization**: nucleation of new grains (used driving force for new grain growth with driving force received from **store energy** in original grains which have dislocation density).
- ❑ **Grain growth**: new grains growth by driving force will be changed to the force at the curvature of the grain boundaries.

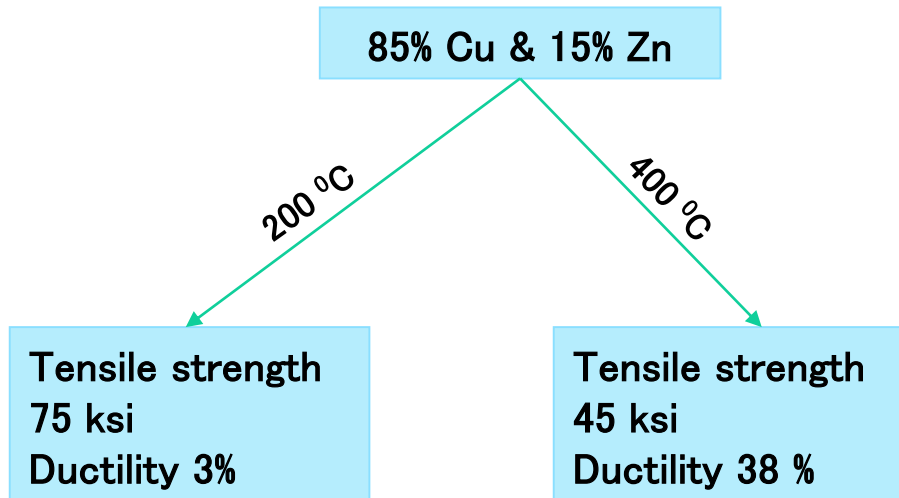
Recovery, Recrystallization and Grain Growth



Recovery, Recrystallization and Grain Growth

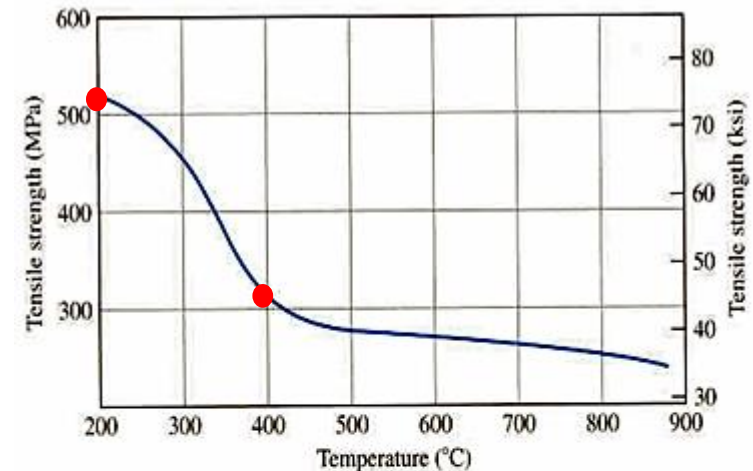
□ Effects on Mechanical Properties

- **Annealing:** decreases tensile strength, increases ductility.
- **Example:**

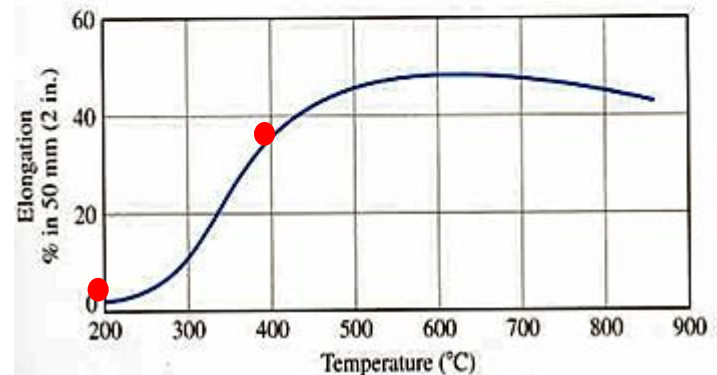


- **Factors affecting recrystallization:**

- Amount of prior deformation
- Temperature and time
- Initial grain size
- Composition of metal



(a)

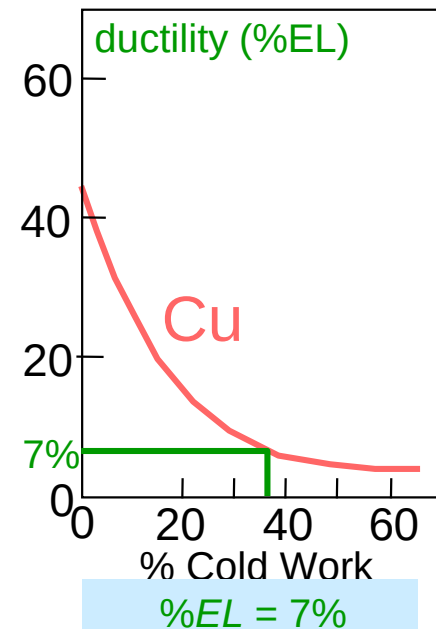
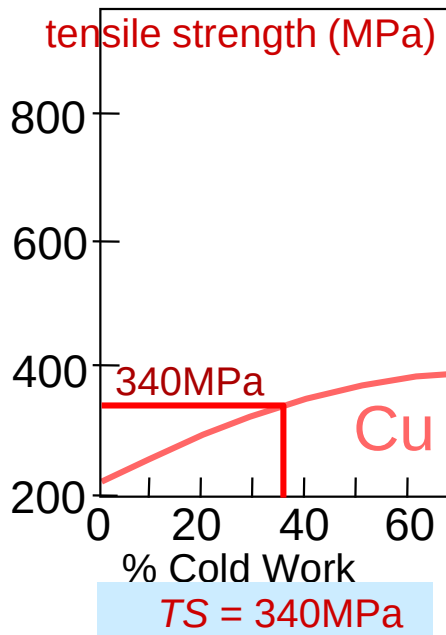
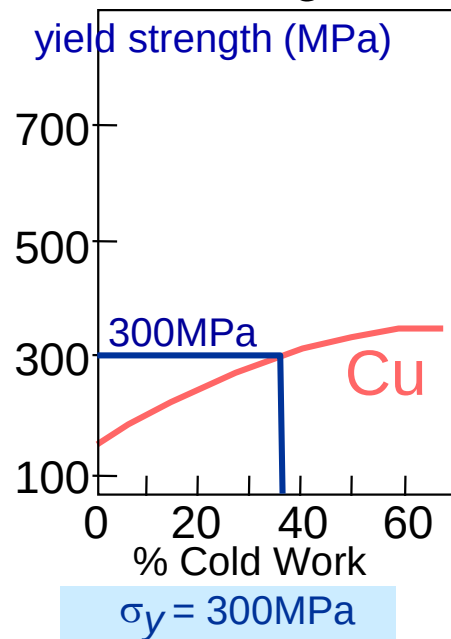
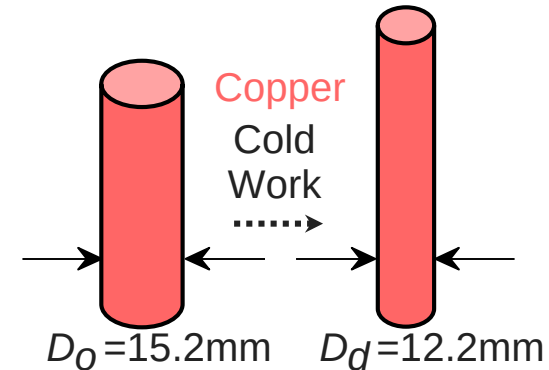


Recovery, Recrystallization and Grain Growth

Effects on Mechanical Properties

- What is the tensile strength & ductility after cold working?

$$\%CW = \frac{\pi r_o^2 - \pi r_d^2}{\pi r_o^2} \times 100 = 35.6\%$$



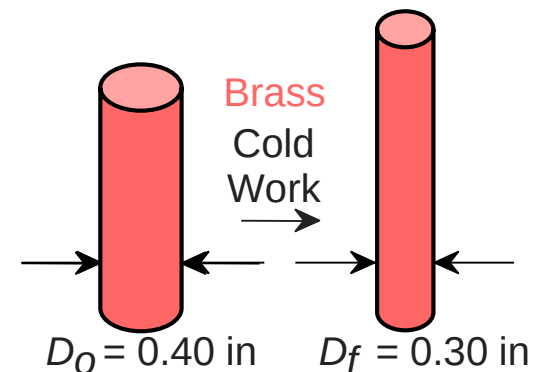
Recovery, Recrystallization and Grain Growth

❑ Cold-working calculations

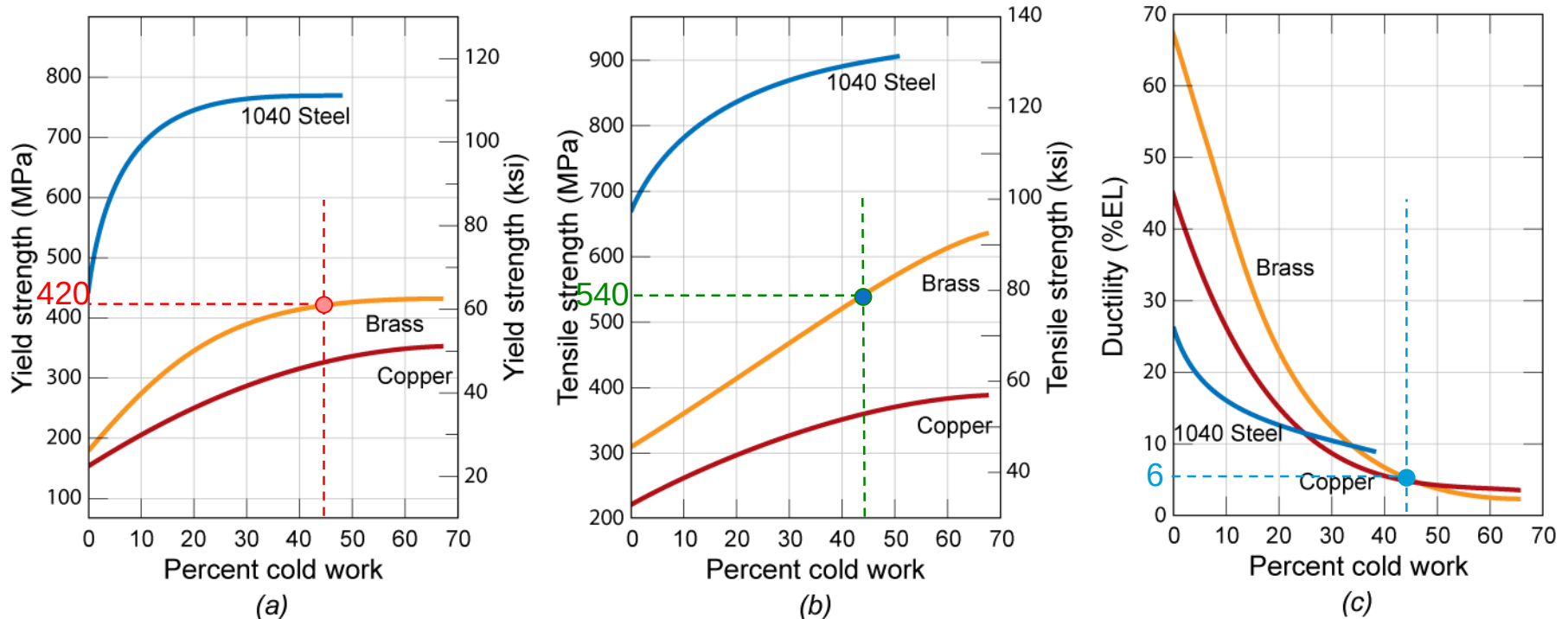
A cylindrical rod of brass originally **0.40 in** in diameter is to be cold worked by drawing. The circular cross section will be maintained during deformation. A cold-worked tensile strength in excess of **380 MPa** and a ductility of at least **15 %EL** are desired. Further more, the final diameter must be **0.30 in**. Explain how this may be accomplished.

- If we directly draw to the final diameter what happens?

$$\begin{aligned}\%CW &= \left(\frac{A_o - A_f}{A_o} \right) \times 100 = \left(1 - \frac{A_f}{A_o} \right) \times 100 \\ &= \left(1 - \frac{\pi D_f^2 / 4}{\pi D_o^2 / 4} \right) \times 100 = \left(1 - \left(\frac{0.30}{0.40} \right)^2 \right) \times 100 = 43.8\%\end{aligned}$$

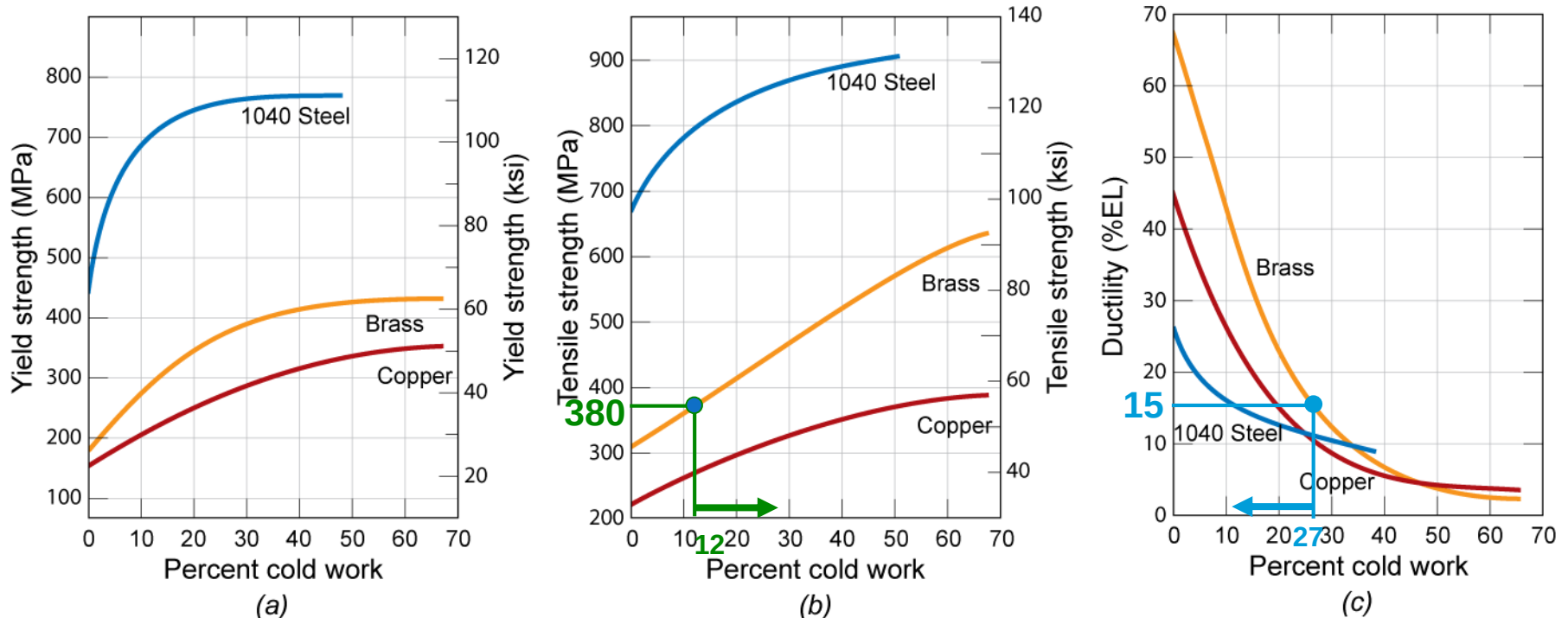


❏ Cold-working calculations



- For % CW = 43.8%
 - $\sigma_y = 420$ MPa
 - $TS = 540$ MPa > 380 MPa
 - % $EL = 6$ < 15 %
- This doesn't satisfy criteria..... what can we do?

❏ Cold-working calculations



For $TS > 380 \text{ MPa}$ $\longrightarrow$ $> 12 \%CW$

For $\%EL < 15$ $\longrightarrow$ $< 27 \%CW$

$\therefore$ our working range is limited to $\%CW = 12-27$

❑ Cold-working calculations

- For objective we need a cold work of $\%CW \approx 12-27$
 - We'll use $\%CW = 20$
- Diameter after first cold draw (before 2nd cold draw)
 - must be calculated as follows:

$$\%CW = \left(1 - \frac{D_{f2}^2}{D_{02}^2} \right) \times 100 \Rightarrow 1 - \frac{D_{f2}^2}{D_{02}^2} = \frac{\%CW}{100}$$

$$\frac{D_{f2}}{D_{02}} = \left(1 - \frac{\%CW}{100} \right)^{0.5} \Rightarrow D_{02} = \frac{D_{f2}}{\left(1 - \frac{\%CW}{100} \right)^{0.5}}$$

$$\therefore \text{Intermediate diameter} = D_{f1} = D_{02} = 0.30 / \left(1 - \frac{20}{100} \right)^{0.5} = 0.335 \text{ in}$$

❑ Cold-working calculations

Summary:

1. Cold work $D_{01} = 0.40 \text{ in} \rightarrow D_{f1} = 0.335 \text{ in}$

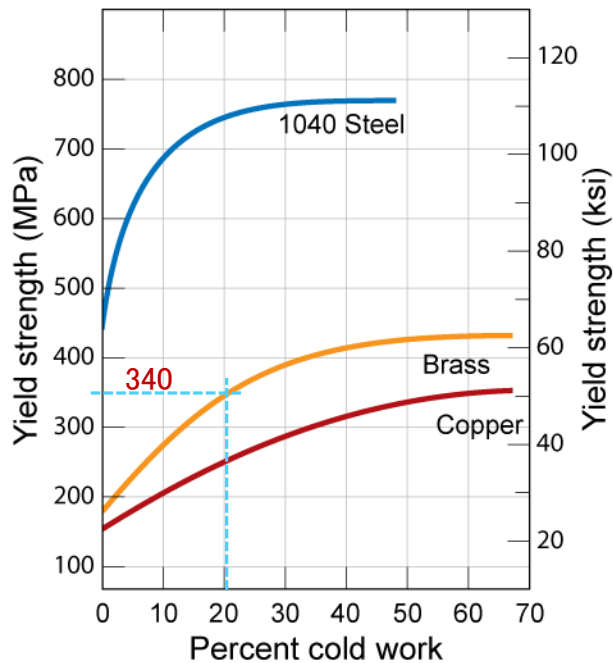
$$\%CW_1 = \left(1 - \left(\frac{0.335}{0.4} \right)^2 \right) \times 100 = 30$$

2. Anneal above $D_{02} = D_{f1}$

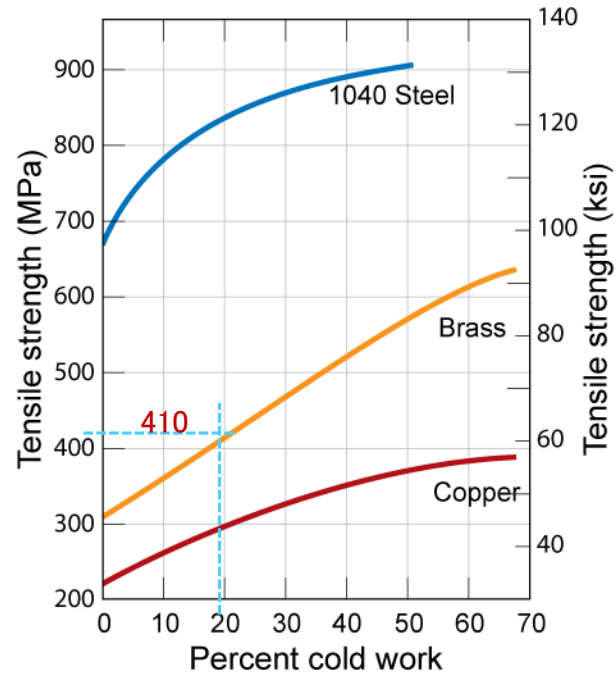
3. Cold work $D_{02} = 0.335 \text{ in} \rightarrow D_{f2} = 0.30 \text{ in}$

$$\%CW_2 = \left(1 - \left(\frac{0.3}{0.335} \right)^2 \right) \times 100 = 20$$

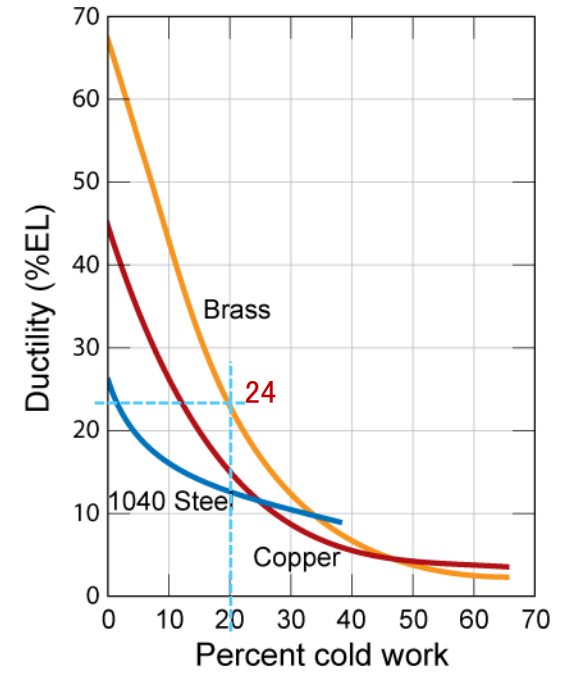
❑ Cold-working calculations



(a)



(b)



(c)

Therefore, meets all requirements →

$$\sigma_y = 340 \text{ MPa}$$

$$TS = 410 \text{ MPa}$$

$$\%EL = 24$$

